Disentangling the Structure of an Antarctic Plankton Food Web in Bloom and Non-bloom Conditions

Master Thesis by

Sarah Lea Mayr

International Master of Science in Marine Biological Resources

At

Ghent University

2025-08

Main Supervisor

Dr. Tomás I. Marina - Centro Austral de Investigaciones Científicas (CADIC-CONICET), Ushuaia, Argentina

Co-supervisor

Dr. Irene R. Schloss - Centro Austral de Investigaciones Científicas (CADIC-CONICET), Ushuaia, Argentina

No data can be taken out of this work without prior approval of the thesis main supervisor.

Index

Executive summary

Potter Cove, Antarctica, has undergone a significant environmental shift over recent decades, transitioning from consistently low phytoplankton biomass to alternating bloom and non-bloom years. The ecological consequences of this drastic change, particularly on the plankton community, but also the wider ecosystem, remain largely unexplored despite phytoplankton's fundamental role. This study addresses this critical gap by investigating the trophic structure and dynamics of the plankton community in Potter Cove under bloom and non-bloom scenarios using a network-based approach. We constructed plankton food webs by compiling Antarctic plankton trophic interactions from a systematic literature review, filtered for species present in Potter Cove. The resulting unweighted food web comprises 37 trophic species and 213 interactions, with a mean trophic level of 1.65 and 41% of omnivore species. Modularity is 0.17 and only the topological roles of Module specialist and Module connector are present.

Our analysis of weighted networks reveals distinct differences between scenarios: connectance (0.19 vs. 0.13), mean trophic level (1.93 vs 1.5), omnivory level (0.06 vs. 0.03), Vulnerability (6.64 vs. 5.9), and Generality (1.62 vs.1.03) are all higher during bloom years. In the bloom year, more species of higher trophic level, including krill, are present, while in the non-bloom year salps and more basal species prevail. There seems to be a positive relationship between a species' interaction strength and trophic level for heterotrophic species and many species that occur in both years have similar interaction strengths. However, there are some species where the interaction strength differs, for example the blooming Large Thalassiosiroids. This research offers crucial insights into energy flow within Antarctic plankton communities and enhances our ability to predict the impacts of environmental change, including climate change, on polar marine ecosystems and their higher trophic levels.

Abstract

Since 2010 the phytoplankton community of Potter Cove, Antarctica, alternates between years of high and low biomass. Before 2010, the phytoplankton biomass was constantly low. Despite phytoplankton's fundamental role in the ecosystem, the trophic dynamics between the planktonic organisms remain largely unexplored. The present study aims to fill this gap by investigating the trophic structure and dynamics of the plankton community in Potter Cove under bloom and non-bloom scenarios using a network-based approach. The assembled food web includes all possible pathways for energy between the species. The comparison of the weighted food web metrics for the bloom and non-bloom scenario showed differences between the two states concerning species composition and interaction strength. In the bloom year more

trophic species of higher trophic level with a greater interaction strength were present indicating a greater availability of food for higher trophic levels. This research shows a more detailed picture of the functioning of the plankton community and builds towards a better understanding of the energy flows within the Antarctic plankton community, enhancing our ability to predict the impacts of environmental change, including climate change, on polar marine ecosystems.

1. Introduction & Aims

1. 1. Importance of studying food webs

Food web analyses are a tool to understand the ecology of a region by looking at the predator-prey (trophic) interactions and therefore at the flow of energy between species. These interactions show dependencies between species and are also a way of transmitting disturbances from one species to another (Rodriguez and Saravia, 2024). Depending on the structure of the food web, the impact of the perturbations can be mitigated or enforced to trophic cascades (Albert et al., 2000). This highlights why it is crucial to study food webs as a way of predicting effects on species resulting from changes like global warming. Furthermore, the study of food webs can give insights into the functioning of a community and lead to conclusions about ecological species roles (e.g. Cirtwill et al., 2018; Dunne et al., 2005; Marina et al., 2024; Riede et al., 2010; Rooney and McCann, 2012).

Contrary to the simple food chain phytoplankton-krill-whales concept that dominated Antarctic ecology some decades ago, it is now clear that many other species play crucial roles in polar food webs (e.g. Hewes et al., 1985; McCormack et al., 2020). One example of this is the food web of Potter Cove, which has been extensively studied (e.g. Cordone et al., 2020; Marina et al., 2018a, 2018b; Rodriguez et al., 2022). Recent studies also include the interaction strength between species, which is defined as the impact of a resource on the consumer (Pawar et al., 2012; Rodriguez and Saravia, 2024). Including measures for the strength, or weight, of the interactions instead of the standard unweighted approach results in more realistic models and offers new insights into the functioning of ecosystems (Kortsch et al., 2021; Rodriguez and Saravia, 2024).

There are many food web studies from Potter Cove (e.g. Cordone et al., 2018; Marina et al., 2018a; Rodriguez et al., 2022), which is an extraordinary case for Antarctica. Furthermore, it was shown that basal species and grazers have a significant impact on interaction strength and stability of the Potter Cove food web (Rodriguez and Saravia, 2024). However, the plankton community is poorly represented, because phytoplankton especially are often grouped into one or few nodes. This simplifies and ignores the interactions between plankton organisms.

1.2. Plankton in Antarctica

Plankton is a highly diverse group of organisms of different sizes, taxonomy, and ecology, comprising both uni- and multicellular organisms. It is classified as either phytoplankton or zooplankton. Phytoplankton are planktonic organisms that synthesize organic compounds via photosynthesis. In contrast, zooplankton are heterotrophic that primarily feed on phytoplankton (Calbet, 2024, chap. 1) and other planktonic organisms. From a food web perspective, organisms producing their own energy are called autotrophs, and organisms without this ability, heterotrophs. However, there are also mixotrophic organisms that are able to produce their own energy, but switch to a heterotrophic mode when the environmental conditions allow it (Stoecker et al., 2017).

Plankton may alternatively be classified according to its size. The resulting groups are macroplankton (2–20 cm), mesoplankton (0.2–20 mm), microplankton (20–200 µm), nanoplankton (2–20 µm), and picoplankton (0.2–2 µm) (e.g. Calbet, 2024, chap. 2; Sieburth et al., 1978). Combining the size categories and feeding mode results in a more detailed classification. For example, small heterotrophic protists, like dinoflagellates, are considered microzooplankton (e.g. Garcia et al., 2016; Sieburth et al., 1978). Some literature suggests further groups such as bacterioplankton and virioplankton which are important for the recycling of nutrients (Calbet, 2024, chap. 1). Another approach is to classify plankton taxonomically, either through morphological methods or through genetic sequencing.

Phytoplankton as a group produces more than half of the oxygen globally (Righetti et al., 2020). This makes phytoplankton one of the most vital groups of organisms in the marine environment, if not the most vital (Calbet, 2024, chap. 1). The phytoplankton community of the Western Antarctic Peninsula is characterized by high seasonality. Coastal phytoplankton in this region is mainly limited by light. Phytoplankton growth starts in spring, caused by increasing day length and retreating sea ice (Smith et al., 2000). Nonetheless, other factors impact phytoplankton growth resulting in great year-to-year variability of phytoplankton biomass (Ducklow et al., 2007; Kim et al., 2018). Phytoplankton maximum abundance in summer is mostly near the surface, where radiation is highest (Smith et al., 1998). Potter Cove (King George/25 de Mayo Island) is not only a representative coastal environment for the Western Antarctic Peninsula but also the site for many studies, and a decade-long phytoplankton monitoring time series is available for this region (e.g. Antoni et al., 2024; Schloss et al., 2012).

Phytoplankton abundance in Potter Cove is historically low compared to other locations (Kim et al., 2018; Schloss et al., 2002). However, in the last decades, there was great interannual variation in the phytoplankton biomass (Schloss et al., 2012). Since 2010, phytoplankton blooms in Potter Cove have both increased significantly and changed their composition (Antoni et al., 2024). Phytoplankton community composition in Potter Cove is dominated by large, centric, micro-sized diatoms. The dominant species are *Porosira glacialis* and *Thalassiosira antarctica* (Hernando et al., 2018; Schloss et al., 2014). Even though these species might only account for 20% of phytoplankton cells, they are found to contain more than 90% of the carbon bound in the phytoplankton (Hernando et al., 2018). However, nanophytoflagellates are the most abundant group when looking at cell counts (Garcia, 2015). Dinoflagellates (mainly *Gyrodinium spp.*) and ciliates (mainly *Strombidinium spp.*) predominate among the microzooplankton while copepods are the most common group of mesozooplankton (Garcia, 2015).

The successful accumulation of a phytoplankton bloom is fundamentally influenced by the physical conditions of the water column, particularly how the depth of particle vertical displacement remains shallower than the critical depth. The critical depth marks where respiration and production of phytoplankton are equal due to lack of light. This condition needs to be sustained for 12-15 days and is crucial for enabling the net growth of phytoplankton by preventing the transport of phytoplankton cells out of the sunlit zone via turbulent mixing (Schloss et al., 2012, 2002). Observations between 2010 and 2020 have indicated a notable overlap of bloom events with years of anomalously hot or cold temperatures (Antoni et al., 2024; Schloss et al., 2012, 2002).

Beyond the physical drivers enabling phytoplankton to bloom, a decoupling of phytoplankton from microzooplankton grazing is equally important for their development. By reducing top-down control by microzooplankton grazers, phytoplankton biomass can accumulate (Garcia et al., 2016). While the predominant perception of a marine system during summer months might suggest a state of stability, even subtle changes can lead to significant shifts within the intricate plankton ecological network (D'Alelio et al., 2015).

The Antarctic Peninsula is one of the regions worldwide where the effects of climate change are most evident, with an increase of the average atmospheric temperature of 1.5° C between 1951 and 2005 (Meredith and King, 2005). This leads to changes in sea ice cover and glacier melting, therefore directly influencing the physical conditions in the water column (e.g. Ducklow et al., 2007; Meredith et al., 2019; Schloss et al., 2014), and hence phytoplankton (Antoni et al., 2024; Hernando et al., 2020; Schloss et al., 2012). Climate change has been shown to greatly affect phytoplankton (Jan et al., 2024). Through trophic interactions, these changes can cascade through the ecosystem to other species. Therefore, it is crucial to understand the dynamics of the food web at the base of the ecosystem.

Many studies have been done worldwide assessing and estimating the impact of changing environmental conditions on plankton (e.g. Garcia et al., 2019; Hernando et al., 2020), but only recently species interactions have been considered (Jan et al., 2024). In some of them, authors could show that changing abiotic conditions lead to a mismatch between phytoplankton and zooplankton spring blooms, creating situations with sparse resources for the higher trophic levels like fish (Jan et al., 2024). However, only few studies investigated this at a high resolution on the plankton level and in a complex (food web) context worldwide (D'Alelio et al., 2019, 2016b, 2016a, 2015; Russo et al., 2023). There are no food web studies focusing on the plankton community available for Antarctica.

The present study aims to disentangle the plankton components of the food web of Potter Cove, by assembling the trophic interactions for the different species in the plankton community, and to estimate the interaction strength to compare between bloom and non-bloom conditions. The hypotheses for this work were to i) show that it is possible to construct a food web model of an Antarctic plankton community, ii) find differences in the weighted food-web metrics and the interaction strength between a bloom and a non-bloom scenario, and iii) gain a more detailed understanding of the functioning of the ecosystem and of the species roles of planktonic organisms.

2. Materials & Methods

2.1. Study area

Potter Cove is one of the best studied areas in Antarctica, with long-term monitoring since the 1970s which created a solid data and knowledge base for the region and made this study possible (e.g. Abele et al., 2017; Antoni et al., 2020; Garcia, 2015; Hernando et al., 2018; Schloss et al., 2002). Potter Cove (Figure 1) is 4 km long and 2.5 km wide fjord-like bay in the southwest of King George / 25 de Mayo Island in the South Shetland Archipelago. It is divided into an inner and an outer section, with different depths and substrates, and with different levels of glacial influence (Schloss et al., 2002).

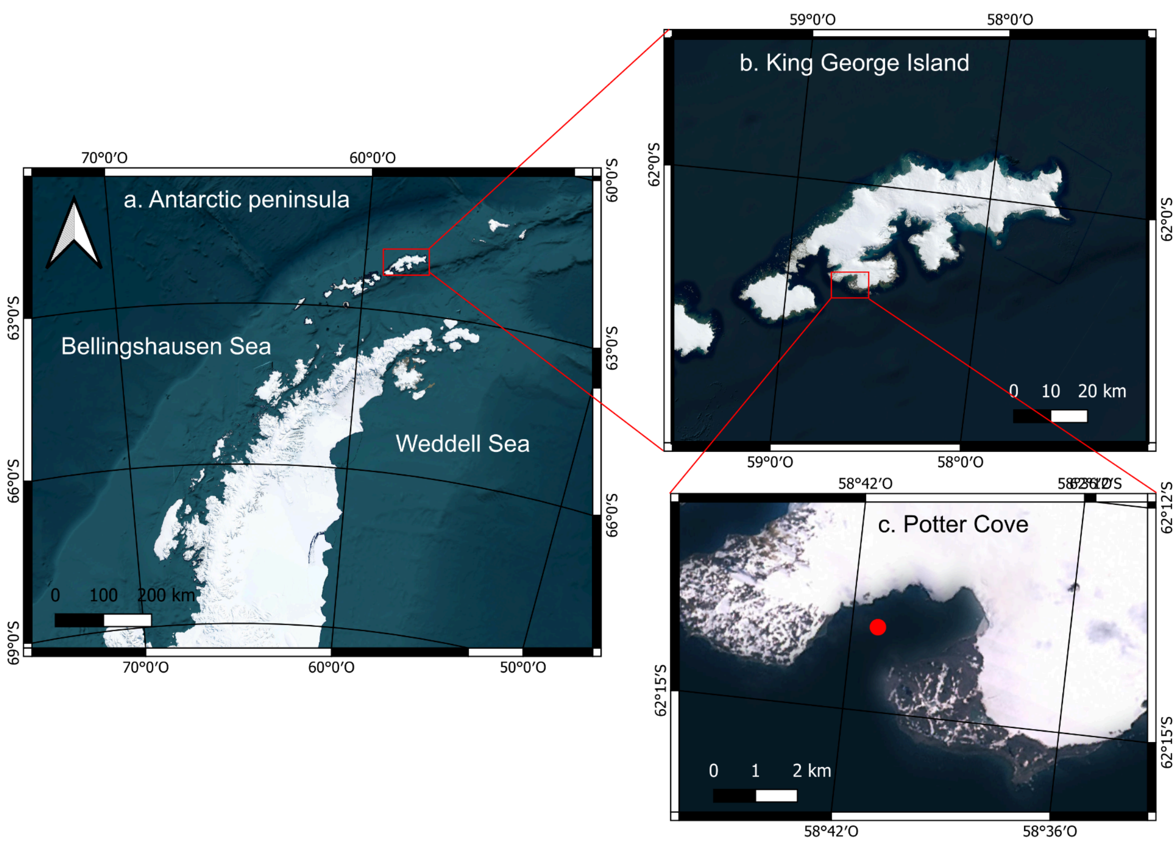

Figure 1: Map of the study area. **A**. The Western Antarctic Peninsula, Antarctica with **b**. King George (25 de Mayo) Island, South Shetlands and **c**. Potter Cove, with the sampling point (red). Map created using QGIS (version 3.44.1).

2.2. Data assemblage

2.2.1. Unweighted metaweb: literature review

A metaweb contains all potential predator-prey interactions between species living in a particular environment and can be viewed as the map of possible energy flow between species.

The plankton food web was assembled with interactions from literature. Starting with the zooplankton species of the existing food web of Potter Cove (Rodriguez et al., 2022), a systematic literature review following the PRISMA approach (O'Dea et al., 2021) was conducted to determine the prey of each zooplankton species (Figure 2). For this the search term "(feeding OR diet* OR (feeding AND strategy) OR prey* OR predator* OR (stomach AND content) OR (grazing AND rate) OR (food AND web) OR trophic) AND ((potter AND cove) OR antarcti*) AND (species name)" was defined and entered in the search engines PubMed (https://pubmed.ncbi.nlm.nih.gov/, 02/2025), Dimensions (https://app.dimensions.ai/discover/publication?search_mode=content, 02/2025), and Scopus (https://www.scopus.com/pages/home?display=basic#basic, 02/2025). Interactions stemmed from observations, stomach and fecal pellet analysis and feeding experiments at the highest possible resolution. A total of 115 papers contained information about interactions and were included in this study. We cleaned and filtered the resulting interactions by matching them to the corresponding observations from Potter Cove. All matches on genus and species level were included. All non-living species were renamed to "detritus". For non-autotrophic species where the literature review did not lead to interactions on high taxonomic resolution, mainly species of the microzooplankton, we used more general information, like size class, after consulting Maximiliano Garcia, a zooplankton Expert at Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET) and the Facultad de Ciencias Veterinarias, Universidad Nacional de La Pampa, Argentina.

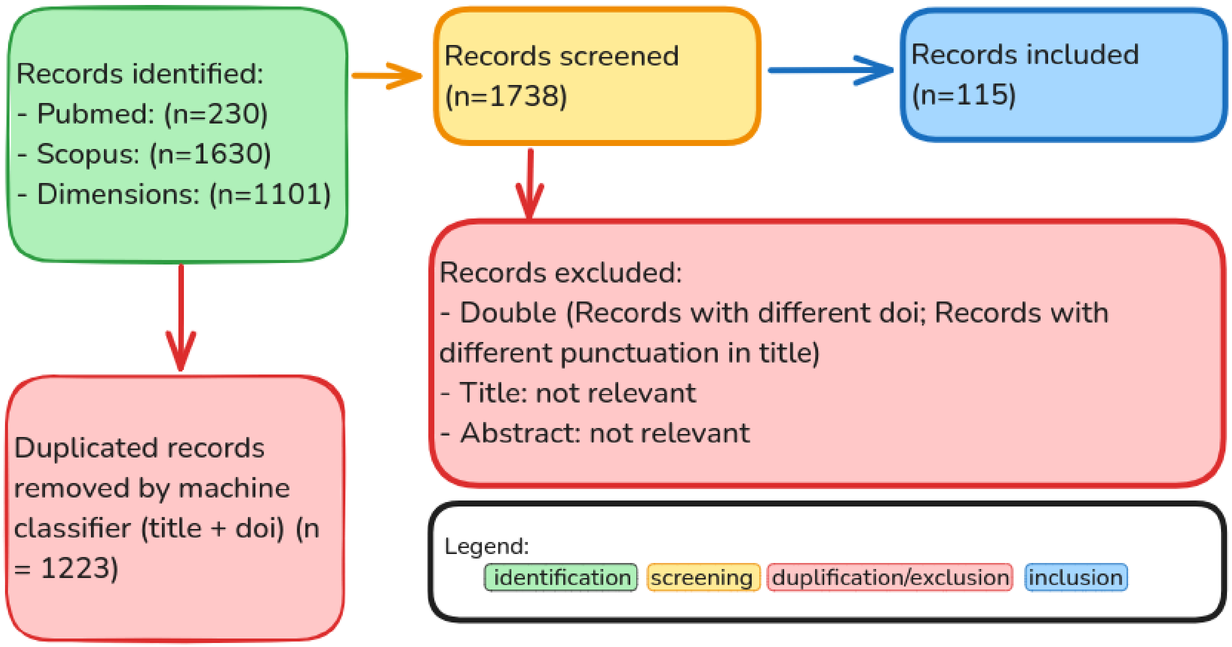

Figure 2: Flowchart of the systematic literature review following the PRISMA approach to find the interactions between the plankton species. A total of 2961 papers were identified from which 1223 could be removed as duplicates. 115 Papers contained interactions used in the analysis.

In this study, only data from the outer Potter Cove (sampling point E2, 62.232 S; 58.688 W) was considered, excluding species only reported from the inner Cove (Antoni et al., 2024). Density data of phytoplankton and zooplankton was used as a seasonal average of the summers of 2014 and 2017. The year 2014 represented a bloom year and 2017 a non-bloom year (Antoni et al., 2024; Garcia, unpublished). For Potter Cove, a bloom condition was defined as the state when Chl-a (a proxy of biomass) concentration was larger than 2 mg m^{-3} (Schloss et al., 2014). Sampling in both years happened between January and March. The seasonal average consists of five data points, where the date of sampling varied between phyto- and zooplankton. The exact day of the sampling varied between phyto- and zooplankton, but the number of samplings stayed the same. Even though the zooplankton data is unpublished, the sampling followed Garcia et al., (2020).

For the food web model, the taxonomic species were aggregated into trophic species to avoid redundancy. A trophic species can be a single taxonomic species or a group of species that occupy the same trophic niche by having the same set of prey and predators (Briand and Cohen, 1984; Rodriguez et al., 2022).

We summarized the species based on their taxonomic group, shape, physiology, the ability to form chains or colonies, and the trophic mode. Especially for the phytoplankton this classification is new and was developed with Gastón Almandoz, phytoplankton expert at the División Ficología, Facultad de Ciencias

Naturales y Museo, Universidad Nacional de La Plata, Argentina. For the classification of the microzooplankton we again worked with Maximiliano Garcia.

2.2.2. Weighted food webs: estimation of interaction strength

The interaction strength was estimated for each pairwise interaction following the methodology from Pawar et al. (2012) as it was recently used for the Potter Cove food web (Rodriguez and Saravia, 2024). The formula for calculating the interaction strength (IS) is:

$$IS = ((a * x_R * m_R)/m_C) \quad \textbf{Equation 1}$$

where a is the dimensionality of the interaction, x_R the density of the resource, m_R the body mass of the resource and m_C the body mass of the consumer (Pawar et al., 2012). Since we considered only interactions from the plankton for this study, both predator and prey move in three dimensions, and therefore the interaction dimensionality stayed the same for all interactions. Density data was obtained from the observations from Potter Cove (Antoni et al., 2024). Body mass as individual wet weight was calculated from biomass and density measurements if available in the observation list (only for a few plankton species). For some zooplankton species, individual body weight in carbon content was available (Garcia et al., 2020) and converted with conversion factors from the literature into individual wet weight. Individual wet weight was available for some species from other food web studies of the region (Rodriguez and Saravia, 2024). For species where no measurement from the Potter Cove area could be found, values from the Weddell Sea database were taken (Jacob et al., 2011). Detritus' "body mass" was estimated as the same as the smallest phytoplankton species (Benthic/ epiphytic). The dataset was filtered by density > 0 for the bloom and non-bloom scenarios, and the weighted food webs for both years were constructed.

Table 1: Table of the weighted and unweighted network metrics used in this study with a definition and ecological relevance

Property	**Definition**	**Ecological relevance**	**Source**
Trophic Species (S)	Total Number of trophic species in the food web	Species from the same ecological niche	(Dunne et al., 2002a)
Predator-Prey Interactions (L)	Total number of directed interactions between the trophic species	Pathways for energy transfer between species	(Dunne et al., 2002a)
Linkage Density (L/S)	Average interactions per species	Information about how connected a species is	(Dunne et al., 2002b)
Connectance (C)	Ratio of realized interactions among possible ones (L/S^2)	Measure of complexity, with implications to robustness	(Dunne et al., 2002b)
Trophic Level (TL)	Distance of the trophic species to the basal level of the food web. $TL_j = 1 + \sum_{i=1}(TL_i P_{ij})$	Measure for efficiency of energy transfer within the web	(Christensen and Pauly, 1992; Rodriguez et al., 2022; Williams and Martinez, 2004)
Percentage of top species	Species with no out-going interactions (out-degree = 0)	Species without predators; top predators	(Briand and Cohen, 1984)
Percentage of intermediate species	Species with in and out degree	Species that act as both, prey and predator	(Briand and Cohen, 1984)
Percentage of basal species	Species with no in-coming interactions (in-degree = 0)	Species at the base of the food web; mainly autotrophic species and detritus	(Briand and Cohen, 1984)
Omnivory level	Diversity of trophic levels in the prey of an omnivore species	If the prey of a species are mostly from the same trophic level or from different ones.	(Christensen and Pauly, 1992)
Omnivory	Percentage of species with prey from different trophic levels	High level of omnivory is associated with higher stability	(Dunne et al., 2002b)
Generality	Average number of prey per species ($(L/S)/fraction_{prey}$)	Variability in the in-degree	(Schoener, 1989)
Vulnerability	Average number of predators per species ($(L/S)/fraction_{predators}$)	Variability in the out-degree	(Schoener, 1989)
Modularity	Calculated with cluster_spinglass algorithm; describes how much the subgroups are connected	Tendancy of the web to form groups; associated with stability since perturbations don't spread to the whole web	(Stouffer and Bascompte, 2011)

2.3. Data Analysis

2.3.1. Unweighted Metaweb: complexity and structure

We calculated food web properties for the assembled unweighted metaweb (Table 1). Food webs are structured around trophic species (S) and predator-prey interactions (L) (Dunne et al., 2002b; Rodriguez et al., 2022). Based on this, we calculated linkage density and connectance (Dunne et al., 2002b).

We estimated the cumulative degree distribution and fitted it to an exponential and a power-law function to test for the best fit (Marina et al., 2018a; Rodriguez et al., 2022). Model selection was based on the Akaike Information Criterion (AIC) fitted to a small sample size.

We calculated mean and maximum trophic level, the percentage of basal, intermediate and top species, omnivory, generality, and vulnerability (Table 1). We used the cluster_spinglass algorithm (Csardi and Nepusz, 2006) to calculate modularity and assigned topological roles (Table 2) (Kortsch et al., 2015). Topological roles were defined based on the standardized within-module degree and the among-module connectivity participation coefficient. The within-module degree is a relative measure of connectance of a species to other species within the same module, and the threshold for this measure is 2.5. The among-module connectivity is a relative measure of the amount of links from a species to other modules; here the threshold value is 0.625 (Kortsch et al., 2015).

Table 2: Names and threshold of the topological roles as defined by (Kortsch et al., 2015)

Name of the topological role	Within-module degree	Among-module Connectivity
Module Hub	> 2.5	< 0.625
Network Connector	> 2.5	> 0.625
Module Specialist	< 2.5	< 0.625
Module Connector	< 2.5	> 0.625

2.3.2. Weighted food webs: bloom and non-bloom comparison at the network and species level

For the comparison of bloom and non-bloom scenarios, the metaweb was filtered by species density within the observation lists for each scenario. For each interaction, the median, standard deviation and mean for 1,000 estimations of the interaction strength were calculated. Since the standard deviation was higher than the mean in many cases, and the distribution of the interaction strength therefore was a right-skewed distribution, the median was used as a more realistic estimate (Rodriguez and Saravia, 2024).

Box plots were used to compare interaction strength and the variation of the interaction strength across the different species and scenarios. Mean trophic level, connectance, linkage density, vulnerability and generality and omnivory level (Table 1), as a representation of the diversity of trophic levels within omnivore species, were calculated as weighted properties for the food webs, as well as the percentage of basal, intermediate and top species.

All analysis have been carried out in R version 4.5.1 (R Core Team, 2025) using the packages igraph 2.1.4 (Csardi and Nepusz, 2006), multiweb 0.8.40.0 (Saravia, 2022), networkExtinction 1.0.3 (Corcoran et al., 2023) and network 1.19.0 (Butts et al., 2024).

3. Results

3.1. Metaweb

The plankton community of Potter Cove consists of 37 trophic species, and the literature review led to 213 predator-prey interactions (Supplementary 1 and 2). The assembled food web (Figure 3) was characterized through a connectance of 0.16 and a linkage density of 5.76. The mean trophic level was 1.65 and the species with the highest trophic level was Chaetognatha (3.36).

Of all species, 41% had prey from more than one trophic level and were therefore considered omnivores. Generality of 12.52 and Vulnerability of 6.09 indicated that each predator may feed on 12 to 13 prey species and each prey species had on average six predators respectively. When looking at the positions of the species within the web, the smallest group was the top species with 5 %. Basal species were the largest group, with more than half (54 %) of all species. Intermediate species, that are prey as well as predators, accounted for the remainder (41 %).

The modularity of the metaweb was 0.17. Since the within module degree of all species was lower than 2.5, two out of the four topological roles (module hubs and network connectors) were not represented by the species of the plankton community. The majority of the species (23) were categorized as module specialists (where among-module connectivity is lower than 0.265) and the remaining 14 species were classified as module connectors (among-module connectivity > 0.265).

The cumulative degree distribution fitted best to an exponential model with an AIC of -93.6 compared to the power-law model with an AIC of -24.1 (Supplementary 4).

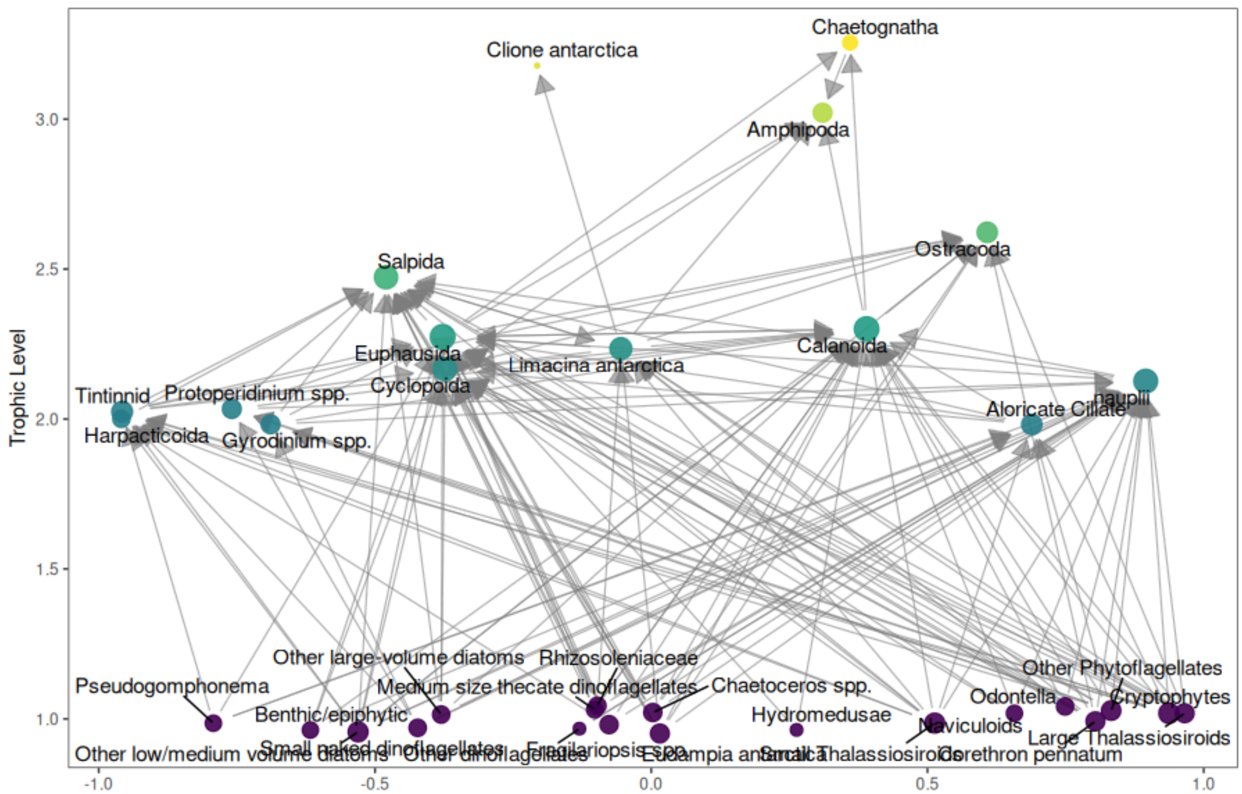

Figure 3: The metaweb of the plankton community of Potter Cove, with 37 trophic species and 213 predator-prey interactions. The size of the points indicates the number of interactions, while the color indicates the trophic level.

3.2. Comparing Weighted Scenarios

The weighted food webs of both scenarios had fewer trophic species than the metaweb, with the bloom food web comprising 22 trophic species and 102 predator-prey interactions (Figure 4A). The non-bloom food web consisted of 26 trophic species and 126 interactions (Figure 4B). Both scenarios shared 17 species, while five species were only present in the bloom year and nine were only present in the non-bloom year. Six species from the metaweb did not occur in either of the scenarios (*Eucampia antarctica*, Medium size thecate Dinoflagellates, *Odontella*, Other large-volume Diatoms, Other low/medium volume Diatoms, Rhizosoleniaceae; Supplementary 2).

The species with the highest trophic level in the bloom scenario were *Clione antarctica* (3.00), and the dinoflagellates *Protoeridinium spp.* (2.98) and *Gyrodinium spp.* (2.98). However, in the non-bloom year, the same dinoflagellates had a trophic level closer to 2 (2.27 and 2.16 respectively), and the species with the highest trophic level was Nauplii (2.59) (Supplementary 6).

Table 3: Weighted food web properties analyzed for the scenarios

Weighted Property	**Bloom**	**Non-bloom**
Linkage density	4.13	3.46
Connectance	0.19	0.13
Maximum trophic level	3.00 (*Clione antarctica*)	2.59 (nauplii)
Mean trophic level	1.93	1.50
Omnivory	55%	42%
Omnivory level	0.06	0.03
Generality	1.62	1.03
Vulnerability	6.64	5.90

The weighted properties in the bloom year had higher values than in the non-bloom year (Table 3). There was also a difference in the percentage of basal and intermediate species between the scenarios. While in the non-bloom year the majority of species were basal (54%), only a bit more than a quarter (27 %) of the species were basal in the bloom scenario. In the latter, more than half were intermediate species (59 %). In the non-bloom scenario, the intermediate species accounted for only 35 %. For both scenarios, the top species were the lowest fraction, 14 % in the bloom and 12 % in the non-bloom year.

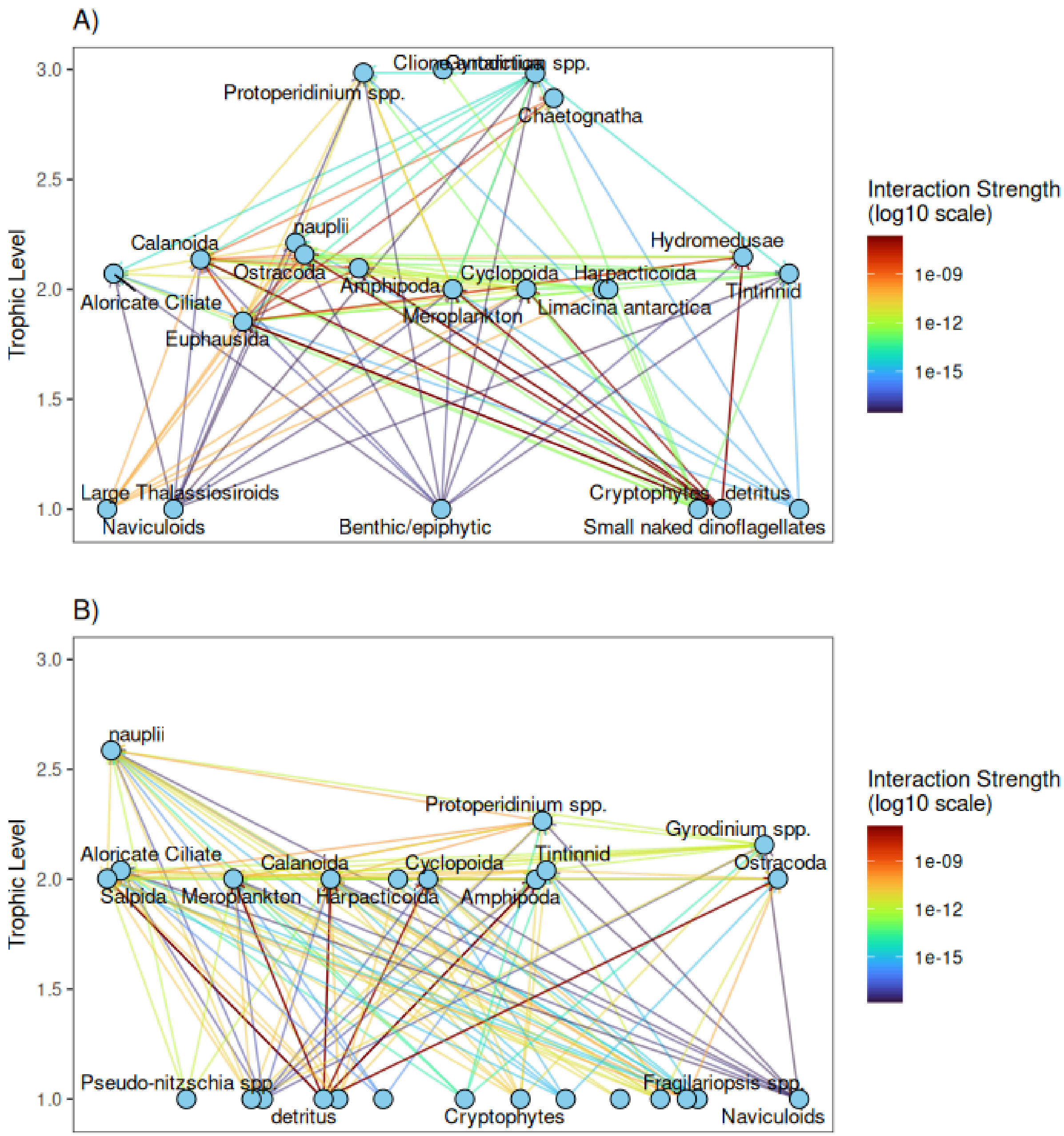

Figure 4: Weighted food webs. **A**) Weighted food web of the bloom year. The food web consists of 22 trophic species and 102 predator-prey interactions. **B**) Weighted non-bloom year food web. The food web consists of 26 trophic species and 126 trophic interactions. The interaction strength is indicated by the color, and detritus has the strongest interactions.

Generality and vulnerability were asymmetrical, with few species concentrating most of the weight of the in- or out-interactions (Figure 5). For vulnerability, which is describing the weight of the out interactions of the prey species, Detritus presented the highest value in both scenarios. In the bloom year it was followed by Euphausida and Amphipoda, and in the non-bloom year by Calanoida and *Protoperidinium spp.* When looking at generality, the species concentrating the most in-interaction strength were Euphausida, Amphipoda, and Calanoida in the bloom year and in the non-bloom year Salpida, Amphipoda, and Calanoida (Supplementary 6).

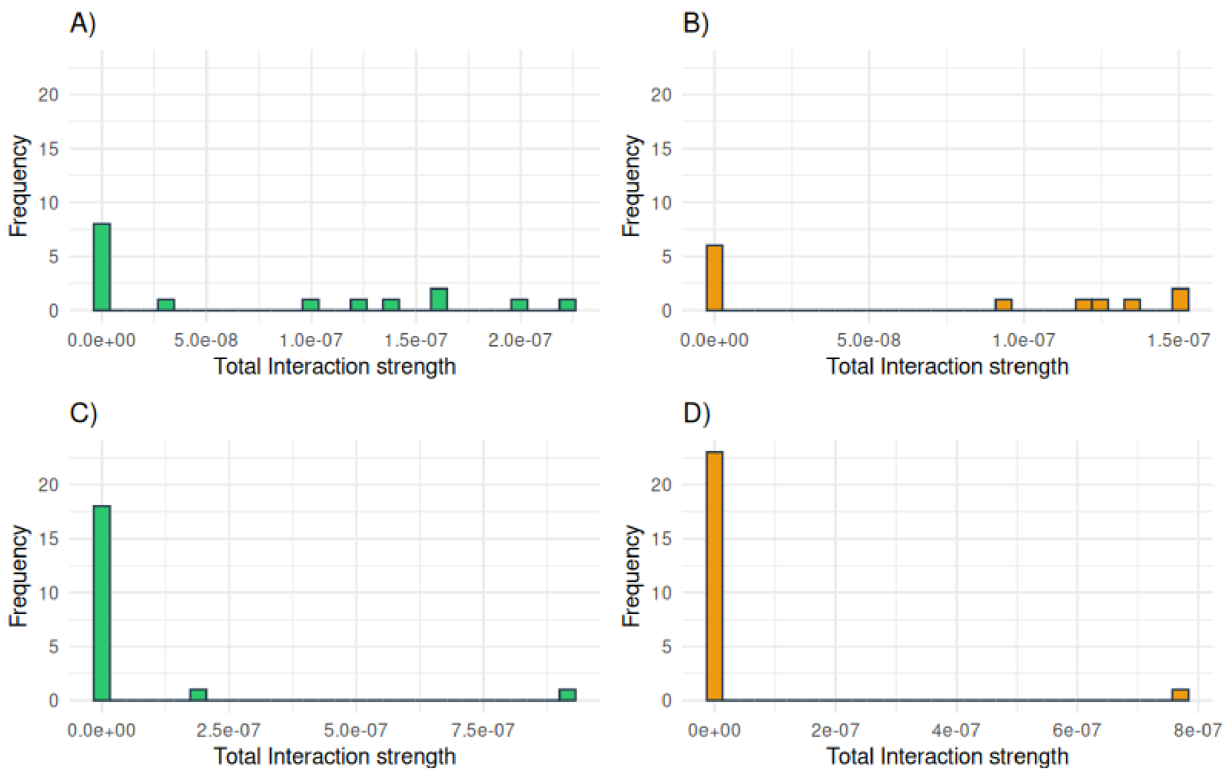

Figure 5: Weighted generality for **A**) the bloom year and **B**) the non-bloom year; and weighted vulnerability for **C**) the bloom year and **D**) the non-bloom year. For generality, the frequency of the interaction strength for all in-interactions is plotted; for vulnerability, the frequency of interaction strength for all out-interactions. The distribution for the four cases is asymmetrical, with only a few species with high interaction strength and many species with low interaction strength.

The analysis of the interaction strength (Figure 6), showed that Detritus displayed the highest interaction strength. A positive relationship between the interaction strength and the trophic level for grazing and predatory species was observed, where a higher trophic level increased the weight of interactions. However, this trend did not include basal species. Furthermore, the strength of the interactions of basal species was less varied, while the predatory species had greater variety in interaction strength. Most species that occurred in both years had interactions of similar weight, but there were a few exceptions. The interaction strength differed for Small naked Dinoflagellates, *Gyrodinium spp.*, Naviculoids, and Large Thalassiosiroids, the blooming species. Figure 6 also illustrates that there were more species of higher trophic level in the bloom year and a higher diversity of basal species in the non-bloom scenario.

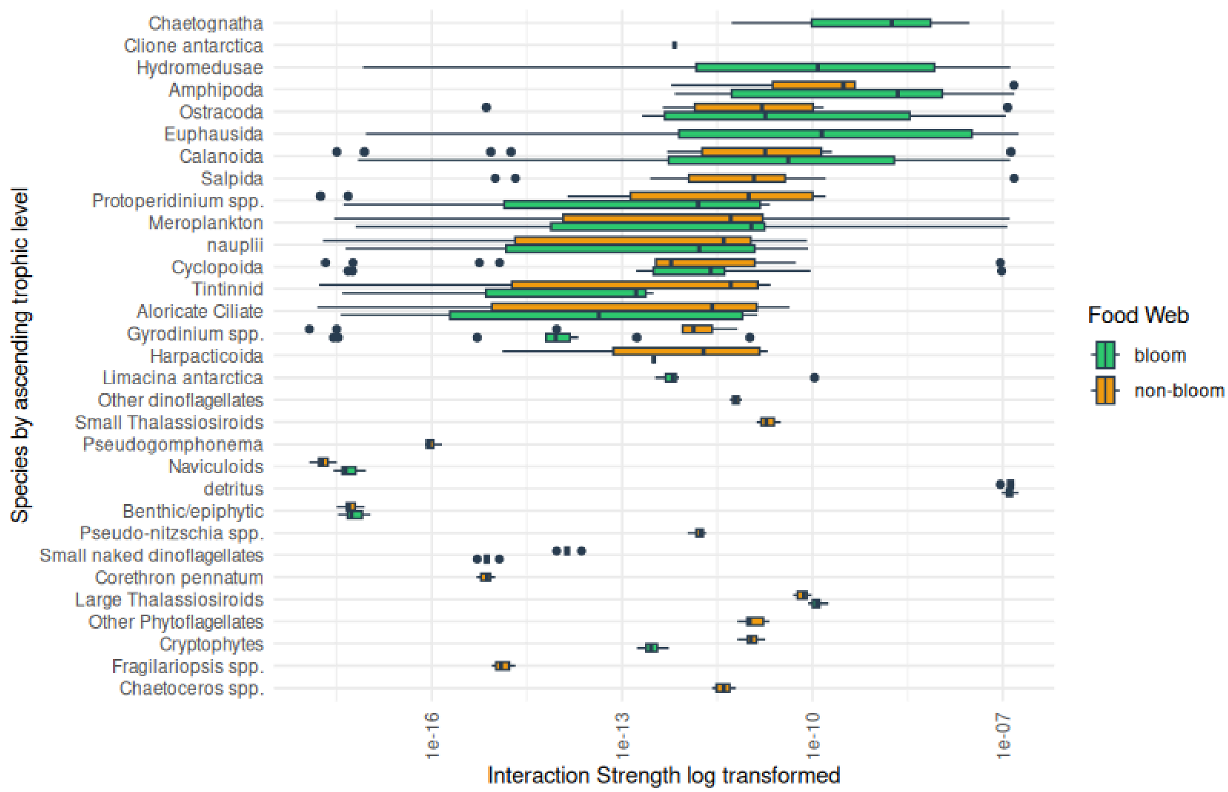

Figure 6: Variation of the interaction strength by species and year. The scale of the interaction strength is logarithmic, and the species are ordered by trophic level. A different behavior of the interaction strength of basal and non-basal species could be observed. The weight of basal interactions did not follow a trend and was also more uniform across interactions for each species, while non-basal species displayed a higher interaction strength with higher trophic levels and greater variability of interaction strength.

4. Discussion

Detailed food web studies focusing on plankton communities are scarce, and none have been conducted in Antarctic ecosystems. Other studies on the Potter Cove food web have been conducted (e.g. Cordone et al., 2018; Marina et al., 2018a; Rodriguez and Saravia, 2024), but since they focused on the whole food web instead of on the plankton, the comparison of the properties is limited. While there are studies of the plankton food web of the Gulf of Naples (Mediterranean Sea), D'Alelio and colleagues (eg. 2016b, 2015) followed a different methodology, focusing on time series, and started with a different database.

4.1.1. Food web metrics of the metaweb mirror the focus on basal species

Connectance, which is a proxy for food web complexity, is lower than what is expected of high-latitude food webs. Typical connectance values of polar food webs range between 0.01 and 0.05 (De Santana et al., 2013). However, D'Alelio et al., (2015) hypothesized based on similar results of high connectance in a plankton food web that low connectance values may result from a low resolution of planktonic and microbial taxa within the food web. Our findings support this interpretation, suggesting that increased taxonomic and trophic resolution may reveal higher levels of connectance in polar systems.

The mean and maximum trophic level (1.65, 3.36) were lower than generally expected of food webs. However, this discrepancy can be explained by the exclusion of higher trophic levels in this study. There are more basal than intermediate species in the food web, even though in most food webs the majority of species are intermediate (Briand and Cohen, 1984). Again, this can be explained by the fact that other food web studies usually do not open the phytoplankton compartment, on one hand, and, on the other, that this study is excluding higher trophic levels.

The percentage of omnivores in the plankton community of Potter Cove was 41%, which is the same as the percentage of intermediate species, and close to the percentage of non-basal species (46%). Generally it is expected to find the percentage of omnivory close to the percentage of intermediate species (Marina et al., 2018a), because there are many interactions between non-basal species.

4.1.2. Distribution of the interactions is asymmetrical among species

The number of interactions, expressed by the cumulative degree distribution, fits best to an exponential function, indicating many species with a low degree, and only a few species with many interactions. It is generally assumed that an asymmetrical cumulative degree distribution makes a food web more vulnerable to the removal of highly connected species, than to the removal of other species. However, because the best fit is an exponential function, and not a power-law function this effect could be stronger (Albert et al., 2000; Marina et al., 2018a). Food webs with an exponential cumulative degree distribution are hypothesized to be more vulnerable to catastrophic fragmentation from random species removal

(Albert et al., 2000). Further stability analysis is necessary to determine if this holds true for the plankton food web, or if it behaves more similar to the full food web of Potter Cove that has been shown to be robust to species loss (Cordone et al., 2020; Rodriguez et al., 2022).

4.1.3. The plankton community is composed of module specialists and module connectors

A modularity of 0.17 is considered to be low. This indicates that the organization of the plankton food web of Potter Cove into sub-networks is not very pronounced. One of the main reasons for modularity seems to be differences in habitat or trophic level (Kortsch et al., 2015). Therefore, the low modularity could be explained by the fact that in this study we are considering only one habitat.

For the topological roles it is important to note that there are different names for the different nodes found in literature, but the thresholds defining them stayed the same (Kortsch et al., 2015; Rodriguez et al., 2022; Russo et al., 2023). All the species are distributed between the two topological roles, module specialist and module connector, without any species being classified as module hub or network connector. The reason for this is related to the low within-module degree of the species, since this measure needs to be above 2.5 for the two absent roles. Network connectors are typically species that are equally connected with all modules, usually by preying on species across modules, as is the case for cod in the Arctic ecosystem (Kortsch et al., 2015). As we did not consider the higher trophic levels here, no network connectors were found; the network connector found at Potter Cove, the black rockcod (*Notothenia coriiceps*), is a top predator (Rodriguez et al., 2022) and was accordingly not included in the food web.

Module hub species are highly connected, but only within their own module. For the plankton food web, the lack of a module hub could be traced back to the generally low modularity, where the species are also connected to other modules. Rodriguez et al., (2022) also found that there are no module hubs in the Potter Cove food web. A third (38%) of all species are module connectors, species that are important for connecting modules with each other but that are not connecting the whole network.

4.2.1 Weighted properties are higher in the bloom scenario

Connectance and Linkage density are lower in the non-bloom scenario than in the bloom one. This could indicate that food webs during bloom conditions have higher robustness against perturbations. However, by themselves they only offer indications, and the tendencies need to be confirmed with other analyses (Marina et al., 2018a; Montoya and Solé, 2003).

Mean and maximum trophic level are higher in the bloom scenario than in the non-bloom scenario. This indicates that the energy transfer in the non-bloom scenario is more efficient than in the bloom scenario, since lower mean trophic levels are associated with more efficient energy transfer and higher stability (Borrelli and Ginzburg, 2014; Cordone et al., 2020). The difference in the mean trophic level cannot only be explained by the higher percentage of basal species in the non-bloom scenario, but also by the species present, generally having a lower trophic level. The only species with a higher trophic level in the non-bloom scenario than in the bloom scenario was Nauplii. This change in trophic level shows that the species can adapt to different conditions of food availability.

Surprisingly, the trophic level of some dinoflagellates was higher than that of Euphausids and Amphipods. This stems from the feeding mode and the high interaction strength of detritus, which is influencing the weighted trophic level.

The shift in species composition, from predominantly basal species in the non-bloom year to a majority of intermediate species in the bloom year, is noteworthy. This contrasts with the general assumption that the ratio of basal to intermediate to top species remains relatively constant (Schoener, 1989). However, it is logical to see the ecological differences reflected in the data, since phytoplankton in a bloom scenario is generally less diverse compared to a non bloom scenario. Studies on the community structure of phytoplankton have shown that single species blooms are frequent for Potter Cove (Antoni et al., 2024). These observations further support the hypothesis of a strong bottom-up effect for the bloom year, as it was already suggested by Antoni and colleagues (2024).

Omnivory is related to the stability of the food web, even though it is primarily a measure of food web structure, not stability (Rodriguez et al., 2022). Omnivorous interactions with a high interaction strength are considered to destabilize the food web, while on the other hand, weak omnivorous interactions are stabilizing (Wootton, 2017). The higher level of omnivory in the bloom scenario would therefore indicate a less stable network during bloom than during non-bloom states (Borrelli and Ginzburg, 2014).

4.2.2 Detritus and high trophic levels showed the greatest interaction strength

Detritus did not only dominate as a prey "species" in the analysis of vulnerability. It also had the highest total interaction strength. This underlines the central role detritus plays in the Potter Cove food web, as already shown by other food web studies in the area (Rodriguez and Saravia, 2024). A central role of detritus is not unusual for marine food webs in general, and emphasizes the importance of dead matter as an energy source (D'Alelio et al., 2016b).

The comparison of the total interaction strength also showed a positive relationship between the trophic level and the interaction strength, where species with higher trophic level have heavier interactions. Basal species, however, are excluded from this observation since in this category there are species with both high and low interaction strength.

The higher trophic level of the species in the bloom scenario leads to the conclusion that there is more energy available for species of higher trophic levels, which were excluded from the present study. The central predators in this scenario, based on the analysis of generality, are Euphausida. In the Southern Ocean, Euphausida - or the main species, krill, *Euphausia superba*, - are considered to be a central prey species to many fish and top predators such as birds and marine mammals (Whitehouse et al., 2009). On the other hand, in the non-bloom scenario Salpida was one of the species with the highest trophic level and replaced krill as the central predator, as the analysis of generality showed. Even though there are some studies suggesting salps, in particular *Sapla thomsoni* may play a role as food for some species, consensus is that in terms of energy salps are a dead end (Pauli et al., 2021). Both species did not co-occur but were only present in one of the scenarios. This is in accordance with the literature, where sometimes even a differentiation between "krill years" and "salp years" is made (Swadling et al., 2023). In this context, krill years are associated with colder water temperature, while salps, more temperate organisms, profit from warmer water temperatures (e.g. Atkinson et al., 2004; Swadling et al., 2023). Since temperature is also influencing bloom formation in Potter Cove (Antoni et al., 2024; Schloss et al., 2012, 2002), it is hard to say what the ultimate cause and what the effect is in this observation. Especially when considering that the bloom year was a relatively colder year and the non-bloom year, a year with average temperatures (Antoni et al., 2024). Further research is necessary to investigate deeper into these dynamics, to verify the implications of the different scenarios on higher trophic levels by integrating the plankton food web into the food web of Potter Cove with higher trophic levels.

Furthermore, it is important to note that the interaction strength as used in the present study is a relative measure of the effect the resource has on the consumer (Pawar et al., 2012). It does not represent an actual energy flow, but only indicates which interactions and species have the greatest impact on the community (Marina et al., 2024; Rodriguez and Saravia, 2024). A more detailed database, with actual measurements of biomass, could add more information about the absolute energy flows and complement the results from this study.

4.3. Limitations and further research

Much of the data used for this study, from species interactions to wet-weight measurements, was obtained from the literature. Although it originated from the Antarctic region, not all of it was specifically from Potter Cove. Using data from this specific region would improve the reliability of the results.

Data availability is also important when deciding on the trophic groups. Bacterioplankton for example is important for nutrient cycling and plankton dynamics (Calbet, 2024, chap. 1). However, abundance data is sparse. It is considered to be more important in winter, when primary production is low and the organisms depend more on recycled organic matter (Rivkin, 1991). Therefore, in this study, which focused on the summer season, when primary production is high, it was assumed that bacterioplankton is only playing a relatively minor role.

Future research could also benefit from comparing the summer food web to other seasons. Seasonality is one of the most defining features in polar regions, and it is well understood that there are great differences between the seasons in the dynamics and even the trophic behavior of many zooplankton species (Garcia et al., 2020). However, because of logistics it is hard to sample in winter and therefore our knowledge about not-summer seasons - especially in Antarctica - is very limited and there is more data from the field season in summer (e.g. Garcia et al., 2020; Schloss et al., 2012).

5. Conclusion

This was the first study on the trophic ecology of an Antarctic plankton community applying a network perspective, and the metaweb gives a map of possibilities of energy flow between the planktonic species of Potter Cove, and could be further used to model plankton food webs in other Antarctic locations. However, it does not display the actual state of the ecosystem. In this sense, adding the interaction strength and focusing on shorter time scenarios gives a more realistic picture.

Adding weighted measures, like interaction strength, is important to understand variability in the functioning and get more detailed results about species roles, as for example the importance of detritus. It is especially useful for the comparison of different states in an ecosystem, where the structure of the food web does not drastically change (Kortsch et al., 2021).

In conclusion, a combination of both weighted and unweighted measures show facets of the ecosystem and do not replace one another, the metaweb showing all possible interactions, and the weighted food webs, presenting actual case studies for contrasting scenarios for which a larger amount of data is necessary.

6. Acknowledgements

I want to take this opportunity to say thank you to my supervisors, Tomás and Irene. You did a wonderful job and without you, this project would not have been possible. I learned a lot under your guidance and am thankful for your council. No matter if it was about the planning of my thesis, various issues I faced along the way, problems with my R code, or me not understanding a paper, I could always come to you to ask, and that was incredible.

Thank you to Maxi, Gastón, Julieta and Iara for making available data for this study, your work was the base this project was built upon. I also want to thank you for the advice you offered, especially concerning the phyto- and microzooplankton. I did not take your help for granted.

Thank you also to my friends, no matter where you are (or better where I have been) you, listened to me, helped me along, and were just generally there to talk, be silly, cook, hike and not get swallowed by thoughts. You are amazing, and I am grateful for every one of you.

Especially thank you to you, Kate, and to you, Romane. You reduced so much of my stress in the last writing phase by offering to (and then actually) proofreading my work. You are great, and I hope to return your kindness!

Ein ganz herzliches Danke geht auch an meine Eltern und meine Familie, vor allem für die Dinge die ich von euch lern, die es an der Uni nicht zu lernen gitb. Aber auch euer Rückhalt bedeuted mir viel, und ohne eure Unterstützung wäre mein Weg garantiert nicht so verlaufen, wie er es ist.

7. References

Abele, D., Vazquez, S., Buma, A.G.J., Hernandez, E., Quiroga, C., Held, C., Frickenhaus, S., Harms, L., Lopez, J.L., Helmke, E., Mac Cormack, W.P., 2017. Pelagic and benthic communities of the Antarctic ecosystem of Potter Cove: Genomics and ecological implications. Mar. Genomics 33, 1–11. https://doi.org/10.1016/j.margen.2017.05.001

Albert, R., Jeong, H., Barabási, A.-L., 2000. Error and attack tolerance of complex networks. Nature 406, 378–382. https://doi.org/10.1038/35019019

Antoni, J.S., Almandoz, G.O., Ferrario, M.E., Hernando, M.P., Varela, D.E., Rozema, P.D., Buma, A.G.J., Paparazzo, F.E., Schloss, I.R., 2020. Response of a natural Antarctic phytoplankton assemblage to changes in temperature and salinity. J. Exp. Mar. Biol. Ecol. 532, 151444. https://doi.org/10.1016/j.jembe.2020.151444

Antoni, J.S., Almandoz, G.O., Goldsmit, J., Garcia, M.D., Flores-Melo, X., Hernando, M.P., Schloss, I.R., 2024. Long-term studies on West Antarctic Peninsula phytoplankton blooms suggest range shifts between temperate and polar species. Glob. Change Biol. 30, e17238. https://doi.org/10.1111/gcb.17238

Atkinson, A., Siegel, V., Pakhomov, E., Rothery, P., 2004. Long-term decline in krill stock and increase in salps within the Southern Ocean. Nature 432, 100–103. https://doi.org/10.1038/nature02996

Borrelli, J.J., Ginzburg, L.R., 2014. Why there are so few trophic levels: Selection against instability explains the pattern. Food Webs 1, 10–17. https://doi.org/10.1016/j.fooweb.2014.11.002

Briand, F., Cohen, J.E., 1984. Community food webs have scale-invariant structure. Nature 307, 264–267. https://doi.org/10.1038/307264a0

Butts, C.T., Hunter, D., Handcock, M., Bender-deMoll, S., Horner, J., Wang, L., Krivitsky, P.N., Knapp, B., Bojanowski, M., Klumb, C., 2024. network: Classes for Relational Data.

Calbet, A., 2024. Plankton in a Changing World: The Impact of Global Change on Marine Ecosystems. Springer Nature Switzerland, Cham. https://doi.org/10.1007/978-3-031-76121-8

Christensen, V., Pauly, D., 1992. ECOPATH II — a software for balancing steady-state ecosystem models and calculating network characteristics. Ecol. Model. 61, 169–185. https://doi.org/10.1016/0304-3800(92)90016-8

Cirtwill, A.R., Dalla Riva, G.V., Gaiarsa, M.P., Bimler, M.D., Cagua, E.F., Coux, C., Dehling, D.M., 2018. A review of species role concepts in food webs. Food Webs 16, e00093. https://doi.org/10.1016/j.fooweb.2018.e00093

Corcoran, D., Ávila-Thieme, M.I., Valdovinos, F.S., Navarrete, S.A., Marquet, P.A., Kusch, E., 2023. NetworkExtinction: Extinction Simulation in Ecological Networks.

Cordone, G., Marina, T.I., Salinas, V., Doyle, S.R., Saravia, L.A., Momo, F.R., 2018. Effects of macroalgae loss in an Antarctic marine food web: applying extinction thresholds to food web studies. PeerJ 6, e5531. https://doi.org/10.7717/peerj.5531

Cordone, G., Salinas, V., Marina, T.I., Doyle, S.R., Pasotti, F., Saravia, L.A., Momo, F.R., 2020. Green vs brown food web: Effects of habitat type on multidimensional stability proxies for a highly-resolved Antarctic food web. Food Webs 25, e00166. https://doi.org/10.1016/j.fooweb.2020.e00166

Csardi, G., Nepusz, T., 2006. The igraph software package for complex network research.

D'Alelio, D., Hay Mele, B., Libralato, S., Ribera d'Alcalà, M., Jordán, F., 2019. Rewiring and indirect effects underpin modularity reshuffling in a marine food web under environmental shifts. Ecol. Evol. 9, 11631–11646. https://doi.org/10.1002/ece3.5641

D'Alelio, D., Libralato, S., Wyatt, T., Ribera d'Alcalà, M., 2016a. Ecological-network models link diversity, structure and function in the plankton food-web. Sci. Rep. 6, 21806. https://doi.org/10.1038/srep21806

D'Alelio, D., Mazzocchi, M.G., Montresor, M., Sarno, D., Zingone, A., Di Capua, I., Franzè, G., Margiotta, F., Saggiomo, V., Ribera d'Alcalà, M., 2015. The green–blue swing: plasticity of plankton food-webs in response to coastal oceanographic dynamics. Mar. Ecol. 36, 1155–1170. https://doi.org/10.1111/maec.12211

D'Alelio, D., Montresor, M., Mazzocchi, M.G., Margiotta, F., Sarno, D., Ribeira d'Alcalà, M., 2016b. Plankton food-webs: to what extent can they be simplified? Adv. Oceanogr. Limnol. 7. https://doi.org/10.4081/aiol.2016.5646

De Santana, C., Rozenfeld, A., Marquet, P., Duarte, C., 2013. Topological properties of polar food webs. Mar. Ecol. Prog. Ser. 474, 15–26. https://doi.org/10.3354/meps10073

Ducklow, H.W., Baker, K., Martinson, D.G., Quetin, L.B., Ross, R.M., Smith, R.C., Stammerjohn, S.E., Vernet, M., Fraser, W., 2007. Marine pelagic ecosystems: the West Antarctic Peninsula. Philos. Trans. R. Soc. B Biol. Sci. 362, 67–94. https://doi.org/10.1098/rstb.2006.1955

Dunne, J.A., Brose, U., Williams, R., Martinez, N., 2005. Modeling food-web dynamics: Complexity-stability implications. pp. 117–129. https://doi.org/10.1093/acprof:oso/9780198564836.003.0011

Dunne, J.A., Williams, R.J., Martinez, N.D., 2002a. Food-web structure and network theory: The role of connectance and size. Proc. Natl. Acad. Sci. 99, 12917–12922. https://doi.org/10.1073/pnas.192407699

Dunne, J.A., Williams, R.J., Martinez, N.D., 2002b. Network structure and biodiversity loss in food webs: robustness increases with connectance. Ecol. Lett. 5, 558–567. https://doi.org/10.1046/j.1461-0248.2002.00354.x

Garcia, M.D., 2015. DINÁMICA ESPACIAL Y TEMPORAL DEL MICRO Y MESOZOOPLANCTON EN CALETA POTTER (ISLA 25 DE MAYO-SHETLAND DEL SUR-ANTÁRTIDA) Y SU RELACIÓN CON EL DERRETIMIENTO GLACIARIO. Universidad nacional del Sur, Bahia Blanka, Argentina.

Garcia, M.D., unpublished. Zooplankton Data Potter Cove.

Garcia, M.D., Dutto, M.S., Chazarreta, C.J., Berasategui, A.A., Schloss, I.R., Hoffmeyer, M.S., 2020. Micro- and mesozooplankton successions in an Antarctic coastal environment during a warm year. PLOS ONE 15, e0232614. https://doi.org/10.1371/journal.pone.0232614

Garcia, M.D., Fernández Severini, M.D., Spetter, C., López Abbate, M.C., Tartara, M.N., Nahuelhual, E.G., Marcovecchio, J.E., Schloss, I.R., Hoffmeyer, M.S., 2019. Effects of glacier melting on the planktonic communities of two Antarctic coastal areas (Potter Cove and Hope Bay) in summer. Reg. Stud. Mar. Sci. 30, 100731. https://doi.org/10.1016/j.rsma.2019.100731

Garcia, M.D., Hoffmeyer, M.S., Abbate, M.C.L., Barría De Cao, M.S., Pettigrosso, R.E., Almandoz, G.O., Hernando, M.P., Schloss, I.R., 2016. Micro- and mesozooplankton responses during two contrasting summers in a coastal Antarctic environment. Polar Biol. 39, 123–137. https://doi.org/10.1007/s00300-015-1678-z

Hernando, M., Schloss, I.R., Almandoz, G.O., Malanga, G., Varela, D.E., De Troch, M., 2018. Combined effects of temperature and salinity on fatty acid content and lipid damage in Antarctic phytoplankton. J. Exp. Mar. Biol. Ecol. 503, 120–128. https://doi.org/10.1016/j.jembe.2018.03.004

Hernando, M., Varela, D.E., Malanga, G., Almandoz, G.O., Schloss, I.R., 2020. Effects of climate-induced changes in temperature and salinity on phytoplankton physiology and stress responses in coastal Antarctica. J. Exp. Mar. Biol. Ecol. 530–531, 151400. https://doi.org/10.1016/j.jembe.2020.151400

Hewes, C.D., Holm-Hansen, O., Sakshaug, E., 1985. Alternate Carbon Pathways at Lower Trophic Levels in the Antarctic Food Web, in: Siegfried, W.R., Condy, P.R., Laws, R.M. (Eds.), Antarctic Nutrient Cycles and Food Webs. Springer, Berlin, Heidelberg, pp. 277–283. https://doi.org/10.1007/978-3-642-82275-9_40

Jacob, U., Thierry, A., Brose, U., Arntz, W.E., Berg, S., Brey, T., Fetzer, I., Jonsson, T., Mintenbeck, K., Möllmann, C., Petchey, O.L., Riede, J.O., Dunne, J.A., 2011. The Role of Body Size in Complex Food Webs: A Cold Case, in: Belgrano, A. (Ed.), Advances in Ecological Research, The Role of Body Size in Multispecies Systems. Academic Press, pp. 181–223. https://doi.org/10.1016/B978-0-12-386475-8.00005-8

Jan, K.M.G., Serandour, B., Walve, J., Winder, M., 2024. Plankton blooms over the annual cycle shape trophic interactions under climate change. Limnol. Oceanogr. Lett. 9, 209–218. https://doi.org/10.1002/lol2.10385

Kim, H., Ducklow, H.W., Abele, D., Ruiz Barlett, E.M., Buma, A.G.J., Meredith, M.P., Rozema, P.D., Schofield, O.M., Venables, H.J., Schloss, I.R., 2018. Inter-decadal variability of phytoplankton biomass along the coastal West Antarctic Peninsula. Philos. Trans. R. Soc. Math. Phys. Eng. Sci. 376, 20170174. https://doi.org/10.1098/rsta.2017.0174

Kortsch, S., Frelat, R., Pecuchet, L., Olivier, P., Putnis, I., Bonsdorff, E., Ojaveer, H., Jurgensone, I., Strāķe, S., Rubene, G., Krūze, Ē., Nordström, M.C., 2021. Disentangling temporal food web dynamics facilitates understanding of ecosystem functioning. J. Anim. Ecol. 90, 1205–1216. https://doi.org/10.1111/1365-2656.13447

Kortsch, S., Primicerio, R., Fossheim, M., Dolgov, A.V., Aschan, M., 2015. Climate change alters the structure of arctic marine food webs due to poleward shifts of boreal generalists. Proc. R. Soc. B Biol. Sci. 282, 20151546. https://doi.org/10.1098/rspb.2015.1546

Marina, T.I., Salinas, V., Cordone, G., Campana, G., Moreira, E., Deregibus, D., Torre, L., Sahade, R., Tatián, M., Barrera Oro, E., De Troch, M., Doyle, S., Quartino, M.L., Saravia, L.A., Momo, F.R., 2018a. The Food Web of Potter Cove (Antarctica): complexity, structure and function. Estuar. Coast. Shelf Sci. 200, 141–151. https://doi.org/10.1016/j.ecss.2017.10.015

Marina, T.I., Saravia, L.A., Cordone, G., Salinas, V., Doyle, S.R., Momo, F.R., 2018b. Architecture of marine food webs: To be or not be a 'small-world.' PLOS ONE 13, e0198217. https://doi.org/10.1371/journal.pone.0198217

Marina, T.I., Saravia, L.A., Kortsch, S., 2024. New insights into the Weddell Sea ecosystem applying a quantitative network approach. Ocean Sci. 20, 141–153. https://doi.org/10.5194/os-20-141-2024

McCormack, S.A., Melbourne-Thomas, J., Trebilco, R., Blanchard, J.L., Constable, A., 2020. Alternative energy pathways in Southern Ocean food webs: Insights from a balanced model of Prydz Bay, Antarctica. Deep Sea Res. Part II Top. Stud. Oceanogr., Ecosystem drivers of food webs on the Kerguelen Axis of the Southern Ocean 174, 104613. https://doi.org/10.1016/j.dsr2.2019.07.001

Meredith, M., Sommerkorn, M., Cassotta, S., Derksen, C., Ekaykin, A., Hollowed, A., Kofinas, G., Mackintosh, A., Melbourne-Thomas, J., Muelbert, M.M.C., Ottensen, G., Prichard, H., Schuur, E.A.G., 2019. Polar Regions, in: IPCC Specieal Report on the Ocean and Cryosphere in a Changing Climate [H.-O. Pörtner, D.C. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, J. Petzold, B. Rama, N.M. Weyer (Eds.)]. Cambridge University Press, pp. 203–320. https://doi.org/10.1017/9781009157964.005

Meredith, M.P., King, J.C., 2005. Rapid climate change in the ocean west of the Antarctic Peninsula during the second half of the 20th century. Geophys. Res. Lett. 32. https://doi.org/10.1029/2005GL024042

Montoya, J.M., Solé, R.V., 2003. Topological properties of food webs: from real data to community assembly models. Oikos 102, 614–622. https://doi.org/10.1034/j.1600-0706.2003.12031.x

O'Dea, R.E., Lagisz, M., Jennions, M.D., Koricheva, J., Noble, D.W.A., Parker, T.H., Gurevitch, J., Page, M.J., Stewart, G., Moher, D., Nakagawa, S., 2021. Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology: a PRISMA extension. Biol. Rev. 96, 1695–1722. https://doi.org/10.1111/brv.12721

Pauli, N.-C., Metfies, K., Pakhomov, E.A., Neuhaus, S., Graeve, M., Wenta, P., Flintrop, C.M., Badewien, T.H., Iversen, M.H., Meyer, B., 2021. Selective feeding in Southern Ocean key grazers—diet composition of krill and salps. Commun. Biol. 4, 1061. https://doi.org/10.1038/s42003-021-02581-5

Pawar, S., Dell, A.I., Van M. Savage, 2012. Dimensionality of consumer search space drives trophic interaction strengths. Nature 486, 485–489. https://doi.org/10.1038/nature11131

QGIS.org, 2025. QGIS Geographic Information System. QGIS Association. http://www.qgis.org

R Core Team, 2025. R: a language and environment for statistical computing (manual). R Foundation for Statistical Computing, Vienna, Austria.

Riede, J.O., Rall, B.C., Banasek-Richter, C., Navarrete, S.A., Wieters, E.A., Emmerson, M.C., Jacob, U., Brose, U., 2010. Chapter 3 - Scaling of Food-Web Properties with Diversity and Complexity Across Ecosystems, in: Woodward, G. (Ed.), Advances in Ecological Research, Ecological Networks. Academic Press, pp. 139–170. https://doi.org/10.1016/B978-0-12-381363-3.00003-4

Righetti, D., Vogt, M., Zimmermann, N.E., Guiry, M.D., Gruber, N., 2020. PhytoBase: A global synthesis of open-ocean phytoplankton occurrences. Earth Syst. Sci. Data 12, 907–933. https://doi.org/10.5194/essd-12-907-2020

Rivkin, R.B., 1991. Seasonal Patterns of Planktonic Production in McMurdo Sound, Antarctica. Am. Zool. 31, 5–16. https://doi.org/10.1093/icb/31.1.5

Rodriguez, I.D., Marina, T.I., Schloss, I.R., Saravia, L.A., 2022. Marine food webs are more complex but less stable in sub-Antarctic (Beagle Channel, Argentina) than in Antarctic (Potter Cove, Antarctic Peninsula) regions. Mar. Environ. Res. 174, 105561. https://doi.org/10.1016/j.marenvres.2022.105561

Rodriguez, I.D., Saravia, L.A., 2024. Potter Cove's Heavyweights: Estimation of Species' Interaction Strength of an Antarctic Food Web. Ecol. Evol. 14, e70389. https://doi.org/10.1002/ece3.70389

Rooney, N., McCann, K.S., 2012. Integrating food web diversity, structure and stability. Trends Ecol. Evol. 27, 40–46. https://doi.org/10.1016/j.tree.2011.09.001

Russo, L., Bellardini, D., Zampicinini, G., Jordán, F., Congestri, R., D'Alelio, D., 2023. From metabarcoding time series to plankton food webs: The hidden role of trophic hierarchy in providing ecological resilience. Mar. Ecol. 44, e12733. https://doi.org/10.1111/maec.12733

Saravia, L.A., 2022. multiweb: ecological network analyses including multiplex networks. R package version 0.6.9.

Schloss, I.R., Abele, D., Moreau, S., Demers, S., Bers, A.V., González, O., Ferreyra, G.A., 2012. Response of phytoplankton dynamics to 19-year (1991–2009) climate trends in Potter Cove (Antarctica). J. Mar. Syst. 92, 53–66. https://doi.org/10.1016/j.jmarsys.2011.10.006

Schloss, I.R., Ferreyra, G.A., Ruiz-Pino, D., 2002. Phytoplankton biomass in Antarctic shelf zones: a conceptual model based on Potter Cove, King George Island. J. Mar. Syst. 36, 129–143. https://doi.org/10.1016/S0924-7963(02)00183-5

Schloss, I.R., Wasilowska, A., Dumont, D., Almandoz, G.O., Hernando, M.P., Michaud-Tremblay, C.-A., Saravia, L., Rzepecki, M., Monien, P., Monien, D., Kopczyńska, E.E., Bers, A.V., Ferreyra, G.A., 2014. On the phytoplankton bloom in coastal waters of southern King George Island (Antarctica) in January 2010: An exceptional feature? Limnol. Oceanogr. 59, 195–210. https://doi.org/10.4319/lo.2014.59.1.0195

Schoener, T.W., 1989. Food Webs from the Small to the Large. Ecology, Mc Arthur Award Lecture 70, 1559–1589.

Sieburth, J.McN., Smetacek, V., Lenz, J., 1978. Pelagic ecosystem structure: Heterotrophic compartments of the plankton and their relationship to plankton size fractions. Limnol. Oceanogr. 23, 1256–1263. https://doi.org/10.4319/lo.1978.23.6.1256

Smith, R.C., Baker, K.S., Vernet, M., 1998. Seasonal and interannual variability of phytoplankton biomass west of the Antarctic Peninsula. J. Mar. Syst. 17, 229–243. https://doi.org/10.1016/S0924-7963(98)00040-2

Smith, W.O., Marra, J., Hiscock, M.R., Barber, R.T., 2000. The seasonal cycle of phytoplankton biomass and primary productivity in the Ross Sea, Antarctica. Deep Sea Res. Part II Top. Stud. Oceanogr., US Southern Ocean JGOFS Program (AESOPS) 47, 3119–3140. https://doi.org/10.1016/S0967-0645(00)00061-8

Stoecker, D.K., Hansen, P.J., Caron, D.A., Mitra, A., 2017. Mixotrophy in the Marine Plankton. Annu. Rev. Mar. Sci. 9, 311–335. https://doi.org/10.1146/annurev-marine-010816-060617

Stouffer, D.B., Bascompte, J., 2011. Compartmentalization increases food-web persistence. Proc. Natl. Acad. Sci. 108, 3648–3652. https://doi.org/10.1073/pnas.1014353108

Swadling, K.M., Constable, A.J., Fraser, A.D., Massom, R.A., Borup, M.D., Ghigliotti, L., Granata, A., Guglielmo, L., Johnston, N.M., Kawaguchi, S., Kennedy, F., Kiko, R., Koubbi, P., Makabe, R., Martin, A., McMinn, A., Moteki, M., Pakhomov, E.A., Peeken, I., Reimer, J., Reid, P., Ryan, K.G., Vacchi, M., Virtue, P., Weldrick, C.K., Wongpan, P., Wotherspoon, S.J., 2023. Biological responses to change in Antarctic sea ice habitats. Front. Ecol. Evol. 10. https://doi.org/10.3389/fevo.2022.1073823

Whitehouse, M.J., Atkinson, A., Ward, P., Korb, R.E., Rothery, P., Fielding, S., 2009. Role of krill versus bottom-up factors in controlling phytoplankton biomass in the northern Antarctic waters of South Georgia. Mar. Ecol. Prog. Ser. 393, 69–82. https://doi.org/10.3354/meps08288

Williams, R.J., Martinez, N.D., 2004. Limits to Trophic Levels and Omnivory in Complex Food Webs: Theory and Data. Am. Nat. 163, 458–468. https://doi.org/10.1086/381964

Wootton, K.L., 2017. Omnivory and stability in freshwater habitats: Does theory match reality? Freshw. Biol. 62, 821–832. https://doi.org/10.1111/fwb.12908

Supplementary Materials

Master Thesis: Disentangling the Structure of an Antarctic Plankton Food Web in Bloom and Non-bloom Conditions

Sarah Mayr

2025-08

The supplementary materials comprise 5 table and 2 figures. The table show the list of interactions (Table 1) and species (Table 2), as well as the results of the food web metrics on the species level for the metaweb (Table 3) and the bloom (Table 4) and non-bloom scenario (Table 5). The figures show the cumulative degree distribution (Figure 1) and the topological roles (Figure 2).

Supplementary 1: Interaction List

Table 1 includes all interactions for the metaweb of Potter Cove with their reference resulting form the systematic literature review. All interactions were validated with by experts on phytoplankton and zooplankton. The interactions from the literature serach were filtered by the assembled groups and for the weighted scenarios. This list was filtered based on density data of the species.

Table 1: List of all predator-prey interactions from the literature used in the study with references

Consumer	Resource	Reference
Calanoida	Aloricate Ciliate	Froneman et al., 1996
Cyclopoida	Aloricate Ciliate	Londsdale et al., 2000
Euphausida	Aloricate Ciliate	Perissinotto et al., 1997
Salpida	Aloricate Ciliate	Froneman et al., 1996
Amphipoda	Amphipoda	Froneman et al., 2000
Calanoida	Benthic/epiphytic	Atkinson 1994
Cyclopoida	Benthic/epiphytic	Atkinson 1994
Euphausida	Benthic/epiphytic	Ligowski 2000
Amphipoda	Calanoida	Pakhomov and Perissinotto 1996
Calanoida	Calanoida	Hopkins 1987
Chaetognatha	Calanoida	Lancraft et al., 1991
Euphausida	Calanoida	Hopkins 1987
Ostracoda	Calanoida	Hopkins 1987
Calanoida	Chaetoceros spp.	Schnack 1985
Cyclopoida	Chaetoceros spp.	Atkinson 1994
Euphausida	Chaetoceros spp.	Froneman et al., 1996
Ostracoda	Chaetoceros spp.	Hopkins 1987
Salpida	Chaetoceros spp.	Montú and Oliveira 1986
Amphipoda	Chaetognatha	Froneman et al., 2000
Calanoida	Corethron pennatum	Yang et al., 2018

(continued)

Consumer	Resource	Reference
Cyclopoida	Corethron pennatum	Atkinson 1994
Euphausida	Corethron pennatum	Hopkins 1985
Ostracoda	Corethron pennatum	Hopkins 1985
Salpida	Corethron pennatum	Hopkins 1985
Calanoida	Cryptophytes	Atkinson 1994
Cyclopoida	Cryptophytes	Atkinson 1994
Euphausida	Cryptophytes	Opaliński et al., 1997
Harpacticoida	Cryptophytes	Swalding et al., 1997
Limacina antarctica	Cryptophytes	Thibodeau et al., 2022
Amphipoda	Cyclopoida	Froneman et al., 2000
Calanoida	Cyclopoida	Hopkins et al., 1993
Cyclopoida	Cyclopoida	Hopkins et al., 1993
Euphausida	Cyclopoida	Hopkins 1987
Ostracoda	Cyclopoida	Hopkins 1985
Salpida	Cyclopoida	Lancraft et al., 1991
Calanoida	Eucampia antarctica	Atkinson 1994
Cyclopoida	Eucampia antarctica	Atkinson 1994
Euphausida	Eucampia antarctica	Schmidt et al., 2006
Ostracoda	Eucampia antarctica	Hopkins 1987
Salpida	Eucampia antarctica	Montú and Oliveira 1986
Amphipoda	Euphausida	Pakhomov and Perissinotto 1996
Calanoida	Euphausida	Hopkins 1985
Chaetognatha	Euphausida	Lancraft et al., 1991
Euphausida	Euphausida	Schmidt et al., 2006
Ostracoda	Euphausida	Hopkins 1987
Salpida	Euphausida	Hopkins 1985
Calanoida	Fragilariopsis spp.	Swalding et al., 1997
Cyclopoida	Fragilariopsis spp.	Swalding et al., 1997
Euphausida	Fragilariopsis spp.	O'Brien et al., 2011
Harpacticoida	Fragilariopsis spp.	Swalding et al., 1997
Salpida	Fragilariopsis spp.	Montú and Oliveira 1986
Calanoida	Gyrodinium spp.	Schnack 1985
Euphausida	Gyrodinium spp.	Schnack 1985
Salpida	Gyrodinium spp.	Montú and Oliveira 1986
Salpida	Harpacticoida	Hopkins et al., 1993
Calanoida	Hydromedusae	Hopkins et al., 1993
Euphausida	Hydromedusae	Cleary et al., 2018
Calanoida	Large Thalassiosiroids	Pasternak & Schnack-Schiel 2001b
Cyclopoida	Large Thalassiosiroids	Hopkins 1987
Euphausida	Large Thalassiosiroids	Schaafsma et al., 2017
Ostracoda	Large Thalassiosiroids	Hopkins 1985
Salpida	Large Thalassiosiroids	Montú and Oliveira 1986
Amphipoda	Limacina antarctica	Froneman et al., 2000
Calanoida	Limacina antarctica	Hopkins 1987
Clione antarctica	Limacina antarctica	Hunt et al., 2008
Cyclopoida	Limacina antarctica	Hopkins 1987
Euphausida	Limacina antarctica	Hopkins 1987

(continued)

Consumer	Resource	Reference
Ostracoda	Limacina antarctica	Hopkins 1987
Salpida	Limacina antarctica	Hopkins et al., 1993
Calanoida	Medium size thecate dinoflagellates	Kruse et al., 2009
Euphausida	Medium size thecate dinoflagellates	Schmidt et al., 2006
Salpida	Medium size thecate dinoflagellates	Montú and Oliveira 1986
Calanoida	Naviculoids	Sarthou et al., 2008
Cyclopoida	Naviculoids	Sarthou et al., 2008
Euphausida	Naviculoids	Meyer and El-Sayed 1983
Calanoida	Odontella	Pasternak & Schnack-Schiel 2001b
Cyclopoida	Odontella	Spinelli et al., 2018
Euphausida	Odontella	Ligowski 2000
Calanoida	Other Phytoflagellates	Pasternak et al., 2009
Euphausida	Other Phytoflagellates	Hopkins et al., 1993
Salpida	Other Phytoflagellates	Hopkins et al., 1993
Calanoida	Other large-volume diatoms	Atkinson 1994
Cyclopoida	Other large-volume diatoms	Atkinson 1994
Euphausida	Other large-volume diatoms	Ligowski 2000
Salpida	Other large-volume diatoms	Montú and Oliveira 1986
Calanoida	Other low/medium volume diatoms	Kruse et al., 2009
Cyclopoida	Other low/medium volume diatoms	Atkinson 1994
Euphausida	Other low/medium volume diatoms	Perissinotto et al., 1997
Harpacticoida	Other low/medium volume diatoms	Swalding et al., 1997
Ostracoda	Other low/medium volume diatoms	Hopkins 1987
Salpida	Other low/medium volume diatoms	Froneman et al., 1996
Calanoida	Protoperidinium spp.	Schnack 1985
Euphausida	Protoperidinium spp.	Froneman et al., 1996
Salpida	Protoperidinium spp.	Froneman et al., 1996
Calanoida	Pseudo-nitzschia spp.	Yang et al., 2018
Euphausida	Pseudo-nitzschia spp.	Passmore et al., 2006
Salpida	Pseudo-nitzschia spp.	Von Harbou et al., 2011
Euphausida	Pseudogomphonema	Ligowski 2000
Calanoida	Rhizosoleniaceae	Schultes et al., 2006
Euphausida	Rhizosoleniaceae	Perissinotto et al., 1997
Salpida	Rhizosoleniaceae	Montú and Oliveira 1986
Calanoida	Small Thalassiosiroids	Schnack 1985
Cyclopoida	Small Thalassiosiroids	Atkinson 1994
Euphausida	Small Thalassiosiroids	Yang et al., 2013
Harpacticoida	Small Thalassiosiroids	Swalding et al., 1997
Salpida	Small Thalassiosiroids	Xue and Zhu 2025
Calanoida	Tintinnid	Hopkins 1987
Cyclopoida	Tintinnid	Hopkins 1987
Euphausida	Tintinnid	Schmidt et al., 2006
Ostracoda	Tintinnid	Hopkins 1987
Salpida	Tintinnid	Montú and Oliveira 1986
Amphipoda	Detritus	Ahn et al., 2021
Calanoida	Detritus	Pasternak et al., 2009
Cyclopoida	Detritus	Hopkins 1985
Euphausida	Detritus	O'Brien 1987

(continued)

Consumer	Resource	Reference
Ostracoda	Detritus	Hopkins 1985
Salpida	Detritus	Hopkins 1985
Calanoida	Nauplii	Pasternak & Schnack-Schiel 2001b
Euphausida	Nauplii	Schmidt et al., 2014
Salpida	Nauplii	Hopkins et al., 1993
Limacina antarctica	Small Thalassiosiroids	Thibodeau et al., 2022
Limacina antarctica	Large Thalassiosiroids	Thibodeau et al., 2022
Limacina antarctica	Rhizosoleniaceae	Thibodeau et al., 2022
Limacina antarctica	Odontella	Thibodeau et al., 2022
Limacina antarctica	Eucampia antarctica	Thibodeau et al., 2022
Limacina antarctica	Corethron pennatum	Thibodeau et al., 2022
Limacina antarctica	Salpida	Thibodeau et al., 2022
Limacina antarctica	Chaetoceros spp.	Thibodeau et al., 2022
Aloricate Ciliate	Chaetoceros spp.	personal communication Maximiliano Garcia
Aloricate Ciliate	Fragilariopsis spp.	personal communication Maximiliano Garcia
Aloricate Ciliate	Naviculoids	personal communication Maximiliano Garcia
Aloricate Ciliate	Benthic/epiphytic	personal communication Maximiliano Garcia
Aloricate Ciliate	Gyrodinium spp.	personal communication Maximiliano Garcia
Aloricate Ciliate	Small Thalassiosiroids	personal communication Maximiliano Garcia
Aloricate Ciliate	Other low/medium volume diatoms	personal communication Maximiliano Garcia
Aloricate Ciliate	Small naked dinoflagellates	personal communication Maximiliano Garcia
Aloricate Ciliate	Cryptophytes	personal communication Maximiliano Garcia
Aloricate Ciliate	Other Phytoflagellates	personal communication Maximiliano Garcia
Aloricate Ciliate	Pseudogomphonema	personal communication Maximiliano Garcia
Tintinnid	Other low/medium volume diatoms	personal communication Maximiliano Garcia
Tintinnid	Small Thalassiosiroids	personal communication Maximiliano Garcia
Tintinnid	Small naked dinoflagellates	personal communication Maximiliano Garcia
Tintinnid	Cryptophytes	personal communication Maximiliano Garcia
Tintinnid	Other Phytoflagellates	personal communication Maximiliano Garcia
Tintinnid	Pseudogomphonema	personal communication Maximiliano Garcia
Tintinnid	Chaetoceros spp.	personal communication Maximiliano Garcia
Tintinnid	Fragilariopsis spp.	personal communication Maximiliano Garcia
Tintinnid	Naviculoids	personal communication Maximiliano Garcia
Tintinnid	Benthic/epiphytic	personal communication Maximiliano Garcia
Tintinnid	Gyrodinium spp.	personal communication Maximiliano Garcia
Nauplii	Small naked dinoflagellates	personal communication Maximiliano Garcia
Nauplii	Medium size thecate dinoflagellates	personal communication Maximiliano Garcia
Nauplii	Gyrodinium spp.	personal communication Maximiliano Garcia
Nauplii	Protoperidinium spp.	personal communication Maximiliano Garcia
Nauplii	Other dinoflagellates	personal communication Maximiliano Garcia
Nauplii	Cryptophytes	personal communication Maximiliano Garcia
Nauplii	Other Phytoflagellates	personal communication Maximiliano Garcia
Nauplii	Pseudogomphonema	personal communication Maximiliano Garcia
Nauplii	Small Thalassiosiroids	personal communication Maximiliano Garcia
Nauplii	Benthic/epiphytic	personal communication Maximiliano Garcia
Nauplii	Naviculoids	personal communication Maximiliano Garcia
Nauplii	Fragilariopsis spp.	personal communication Maximiliano Garcia

(continued)

Consumer	Resource	Reference
Nauplii	Chaetoceros spp.	personal communication Maximiliano Garcia
Nauplii	Odontella	personal communication Maximiliano Garcia
Nauplii	Other large-volume diatoms	personal communication Maximiliano Garcia
Nauplii	Rhizosoleniaceae	personal communication Maximiliano Garcia
Nauplii	Pseudo-nitzschia spp.	personal communication Maximiliano Garcia
Nauplii	Corethron pennatum	personal communication Maximiliano Garcia
Nauplii	Eucampia antarctica	personal communication Maximiliano Garcia
Nauplii	Large Thalassiosiroids	personal communication Maximiliano Garcia
Nauplii	Other low/medium volume diatoms	personal communication Maximiliano Garcia
Gyrodinium spp.	Small naked dinoflagellates	personal communication Maximiliano Garcia
Gyrodinium spp.	Cryptophytes	personal communication Maximiliano Garcia
Gyrodinium spp.	Other Phytoflagellates	personal communication Maximiliano Garcia
Gyrodinium spp.	Naviculoids	personal communication Maximiliano Garcia
Gyrodinium spp.	Benthic/epiphytic	personal communication Maximiliano Garcia
Gyrodinium spp.	Gyrodinium spp.	personal communication Maximiliano Garcia
Protoperidinium spp.	Small naked dinoflagellates	personal communication Maximiliano Garcia
Protoperidinium spp.	Cryptophytes	personal communication Maximiliano Garcia
Protoperidinium spp.	Other Phytoflagellates	personal communication Maximiliano Garcia
Protoperidinium spp.	Naviculoids	personal communication Maximiliano Garcia
Protoperidinium spp.	Benthic/epiphytic	personal communication Maximiliano Garcia
Protoperidinium spp.	Gyrodinium spp.	personal communication Maximiliano Garcia
Hydromedusae	Benthic/epiphytic	Rodriguez et el., 2024
Hydromedusae	Ostracoda	Rodriguez et el., 2024
Hydromedusae	Detritus	Rodriguez et el., 2024
Hydromedusae	Euphausida	Rodriguez et el., 2024
Amphipoda	Meroplankton	Sheader and Evans 1975
Gyrodinium spp.	Meroplankton	Johnson and Shanks 2003
Protoperidinium spp.	Meroplankton	Johnson and Shanks 2003
Meroplankton	Nauplii	Harvey and Morrier 2003
Meroplankton	Small naked dinoflagellates	Fileman et al., 2014
Meroplankton	Medium size thecate dinoflagellates	Fileman et al., 2014
Meroplankton	Gyrodinium spp.	Fileman et al., 2014
Meroplankton	Protoperidinium spp.	Fileman et al., 2014
Meroplankton	Other dinoflagellates	Fileman et al., 2014
Meroplankton	Cryptophytes	Ameneiro et al., 2016
Meroplankton	Other Phytoflagellates	Fileman et al., 2014
Meroplankton	Pseudogomphonema	Tanković et al., 2018
Meroplankton	Small Thalassiosiroids	Tanković et al., 2018
Meroplankton	Benthic/epiphytic	Tanković et al., 2018
Meroplankton	Naviculoids	Tanković et al., 2018
Meroplankton	Fragilariopsis spp.	Tanković et al., 2018
Meroplankton	Chaetoceros spp.	Tanković et al., 2018
Meroplankton	Odontella	Tanković et al., 2018
Meroplankton	Other large-volume diatoms	Tanković et al., 2018
Meroplankton	Rhizosoleniaceae	Tanković et al., 2018
Meroplankton	Pseudo-nitzschia spp.	Tanković et al., 2018
Meroplankton	Corethron pennatum	Tanković et al., 2018
Meroplankton	Eucampia antarctica	Tanković et al., 2018

(continued)

Consumer	Resource	Reference
Meroplankton	Large Thalassiosiroids	Tanković et al., 2018
Meroplankton	Other low/medium volume diatoms	Tanković et al., 2018
Meroplankton	Detritus	Carrier and Reitzel 2020

Supplementary 2: Species List

Table 2 includes all trophic species reported for station E2 in Potter Cove. It also includes for which scenario the species was reported. "Bloom" and "non-bloom" mean, that the species was only present in one of the scenarios, "both" indicates that the species was present in both scenarios and "non" indicates, that the species was not in any scenario, but is generally reported from Potter Cove and was therefore included in the metaweb.

Table 2: Trophic species used in this study, the description of the group, and with the species on the highest taxonomic level they were obsereved in Potter Cove

Trophic Species	Scenario	Description	Observed Species
Aloricate Ciliate	both	Microzooplankton	myrionecta; Aloricate Ciliate; didinium gargantua; leegaardiella elbraechteri; lohmanniella oviformis; monodinium balbianii; pelagostrobilidium neptuni; peritrichia; rimostrombidium glacicolum; strombidium; rimostrombidium conicum
Amphipoda	both	zooplankton	Themisto gaudichaudii
Benthic/epiphytic	both	Pennated diatoms, several benthic or epiphytic genus/species with diverse shapes and sizes. Usually in low numbers.	Licmophora; Cocconeis; pleurosigma; cocconeis; amphora; achnanthes
Calanoida	both	Zooplankton, group of copepods	Calanoides acutus; Calanus propinquus; Ctenocalanus citer; Metridia gerlachei; rhincalanus gigas
Chaetoceros spp.	non-bloom	Diverse genus of spiny centric diatoms. Chain forming. Bloom forming.	Chaetoceros; Chaetoceros atlanticus; Chaetoceros neogracilis; Chaetoceros flexuosus; Chaetoceros minimus; Chaetoceros concavicornis
Chaetognatha	bloom	Zooplankton	Sagitta
Clione antarctica	bloom	Zooplankton, Pteropoda	Clione antarctica
Corethron pennatum	non-bloom	Big centric diatom. Solitary. Bloom forming.	Corethron pennatum
Cryptophytes	both	Includes cryptophytes, a group of nanoflagellates that usually form blooms in Antarctica	Cryptophyceae
Cyclopoida	both	Zooplankton, group of copepods	Oithona similis; Oithona frigida; Oncaea curvata
Eucampia antarctica	non	Big centric diatom, usually in low low cell numbers. Can form chains.	Eucampia antarctica
Euphausida	bloom	Zooplankton	Euphausia superba

(continued)

Trophic Species	Scenario	Description	Observed Species
Fragilariopsis spp.	non-bloom	Several chain forming species (F. kerguelensis, F. curta, F. rhombica) and smaller species (e.g. F. nana, F. pseudonana). Some reach high abundances.	Fragilariopsis; fragilariopsis; Fragilariopsis rhombica
Gyrodinium spp.	both	Dinoflagellates, several Gyrodinium species, usually without cloroplasts.	Gyrodinium
Harpacticoida	both	Zooplankton, group of copepods	Harpacticoida
Hydromedusae	bloom	Zooplankton	Hydromedusae
Large Thalassiosiroids	both	Centric bigger diatoms (>20 um), including several genera. Solitary or in chains. Bloom forming.	Thalassiosira > 20 µm; Porosira glacialis; Coscinodiscus
Limacina antarctica	bloom	Zooplankton, Pteropoda	Limacina antarctica
Medium size thecate dinoflagellates	non	Several genus/species of thecate dinoflagellates, mostly medium in size and with cloroplasts.	prorocentrum; peridiniella; dinophysis
Meroplankton	both	Zooplankton, larval stages of benthic animals	annelida; gastropoda; decapoda; echinodermata
Naviculoids	both	Several unidentified taxa with a 'naviculoid shape'. Usually in low numbers.	naviculaceae; navicula directa
Odontella	non	The most common O. weissfloggi, usually form blooms with high biomass values (large volume species)	Odontella
Ostracoda	both	Zooplankton	Conchoecia
Other Phytoflagellates	non-bloom	Includes other phytoflagellates grouped together to simplify data.	Prasinophyceae; silicoflagellida
Other dinoflagellates	non-bloom	Other naked/thecate dinos of 'medium/large' size grouped together to simplify data. Usually in low numbers.	oxytoxum turbo; oxytoxum ovum; torodinium; warnowia
Other large-volume diatoms	non	Several genus/species with large volumes. Grouped together to simplify data. Usually in low numbers. Thalassiothrix antarctica has extremely large cells.	Manguinea; thalassiothrix; grammatophora; Plagiotropis
Other low/medium volume diatoms	non	Several genus/species with low volumes, Grouped together to simplify data. Usually in low numbers.	cylindrotheca; nitzschia; fragilaria

(continued)

Trophic Species	Scenario	Description	Observed Species
Protoperidinium spp.	both	Dinoflagellates, several Protoperidinium and 'similar' taxa, usually without cloroplasts.	protoperidinium; preperidinium meunierii
Pseudo-nitzschia spp.	non-bloom	Several chain forming species with acute cells. Some reach high abundances.	Pseudo-nitzschia
Pseudogomphonema	non-bloom	The most common species is P. kamtschaticum, a small species that can ocassionally reach high abundances.	Pseudogomphonema kamtschaticum
Rhizosoleniaceae	non	Large centric diatoms with long cilindrical shape. Can form chains. Usually in low cell numbers.	guinardia; rhizosolenia; proboscia
Salpida	non-bloom	Zooplankton	Salpa thompsoni
Small Thalassiosiroids	non-bloom	Centric small diatoms (<20 um) inlcuiding several Thalassiosira and Shionodiscus species. Solitary or in chains. Bloom forming.	Shionodiscus gaarderae; Thalassiosira $<$ 20 µm
Small naked dinoflagellates	both	Several genus/species of naked dinoflagellates, mostly small in size and with cloroplasts.	Gymnodinium; katodinium; karenia; lepidodinium
Tintinnid	both	Microzooplankton, Ciliates with lorica	Tintinnid; codonellopsis glacialis; codonellopsis balechi; cymatocylis convallaria
Detritus	both	Dead organic matter	detritus
Nauplii	both	Microzooplankto, larval states of crustaceans	nauplii

Supplementary 3: Results of the Metaweb

Table 3 consists of the results on the species level for the metaweb and includes the trophic species (Species), the trophic level (TL), the Omnivory index (OI), the total Degree (Tot_Deg), the in- and out-degree (In_Deg, Out_Deg) as well as the topological Role (Role). Column “Module” indicates the module the species is part of and within-module degree (WMD) and among-module connectivity (AMC) were used to calculate the topological roles.

Table 3: Results of the metaweb ordered by trophic level

Species	TL	OI	Tot_Deg	In_Deg	Out_Deg	Role	Module	WMD	AMC
Chaetognatha	3.36	0.00	3	2	1	modspe	1	-0.96	0.00
Clione antarctica	3.16	0.00	1	1	0	modspe	1	-1.38	0.00
Amphipoda	2.94	0.55	9	8	1	modspe	1	0.08	0.20
Hydromedusae	2.73	0.54	6	4	2	modspe	1	-0.54	0.28
Ostracoda	2.56	0.38	12	11	1	modspe	1	0.50	0.29
Salpida	2.47	0.32	23	22	1	modcon	2	1.63	0.65
Calanoida	2.37	0.38	32	27	5	modspe	1	1.76	0.62
Euphausida	2.36	0.37	35	28	7	modcon	1	1.97	0.63
Protoperidinium spp.	2.31	0.24	12	7	5	modspe	3	0.42	0.61
Limacina antarctica	2.16	0.21	16	9	7	modspe	1	0.71	0.46
Meroplankton	2.15	0.15	26	23	3	modcon	2	1.63	0.66
Cyclopoida	2.14	0.24	23	17	6	modspe	1	1.13	0.59
Nauplii	2.11	0.12	25	21	4	modcon	2	2.00	0.66
Aloricate Ciliate	2.09	0.09	15	11	4	modcon	3	0.78	0.64
Tintinnid	2.09	0.09	16	11	5	modcon	3	0.81	0.65
Gyrodinium spp.	2.02	0.33	16	7	9	modspe	3	1.99	0.53
Harpacticoida	2.00	0.00	5	4	1	modspe	2	-0.22	0.32
Benthic/epiphytic	1.00	0.00	10	0	10	modcon	3	-0.40	0.64
Chaetoceros spp.	1.00	0.00	10	0	10	modspe	1	-0.54	0.62
Corethron pennatum	1.00	0.00	8	0	8	modspe	1	-0.54	0.47
Cryptophytes	1.00	0.00	11	0	11	modcon	3	-0.40	0.66
Eucampia antarctica	1.00	0.00	8	0	8	modspe	1	-0.54	0.47
Fragilariopsis spp.	1.00	0.00	9	0	9	modcon	2	-0.22	0.64
Large Thalassiosiroids	1.00	0.00	8	0	8	modspe	1	-0.54	0.47
Medium size thecate dinoflagellates	1.00	0.00	5	0	5	modspe	2	-0.59	0.48
Naviculoids	1.00	0.00	9	0	9	modcon	3	-0.37	0.64
Odontella	1.00	0.00	6	0	6	modspe	1	-0.75	0.44
Other Phytoflagellates	1.00	0.00	9	0	9	modcon	3	-0.40	0.64

(continued)

Species	TL	OI	Tot_Deg	In_Deg	Out_Deg	Role	Module	WMD	AMC
Other large-volume diatoms	1.00	0.00	6	0	6	modspe	2	-0.59	0.50
Other low/medium volume diatoms	1.00	0.00	10	0	10	modcon	2	-0.22	0.64
Pseudo-nitzschia spp.	1.00	0.00	5	0	5	modspe	2	-0.59	0.48
Pseudogomphonema	1.00	0.00	5	0	5	modcon	3	-1.15	0.64
Rhizosoleniaceae	1.00	0.00	6	0	6	modspe	2	-0.59	0.50
Small Thalassiosiroids	1.00	0.00	10	0	10	modcon	2	-0.22	0.64
Detritus	1.00	0.00	8	0	8	modspe	1	-0.33	0.38
Small naked dinoflagellates	1.00	0.00	6	0	6	modspe	3	-0.37	0.44
Other dinoflagellates	1.00	0.00	2	0	2	modspe	2	-0.97	0.00

Supplementary 4: Degree distribution of the Metaweb

The cumulative degree distribution was fitted to an exponential and a power-law model to test for the best fit (Corcoran et al., 2023). Model selection was based on the Akaike Information Criterion (AIC) fitted to a small sample size.
The best fit for the cumulative degree distribution was the exponential function with an AIC of -93.6 compared to the power-law function with an AIC of -24.1 (Figure 1).

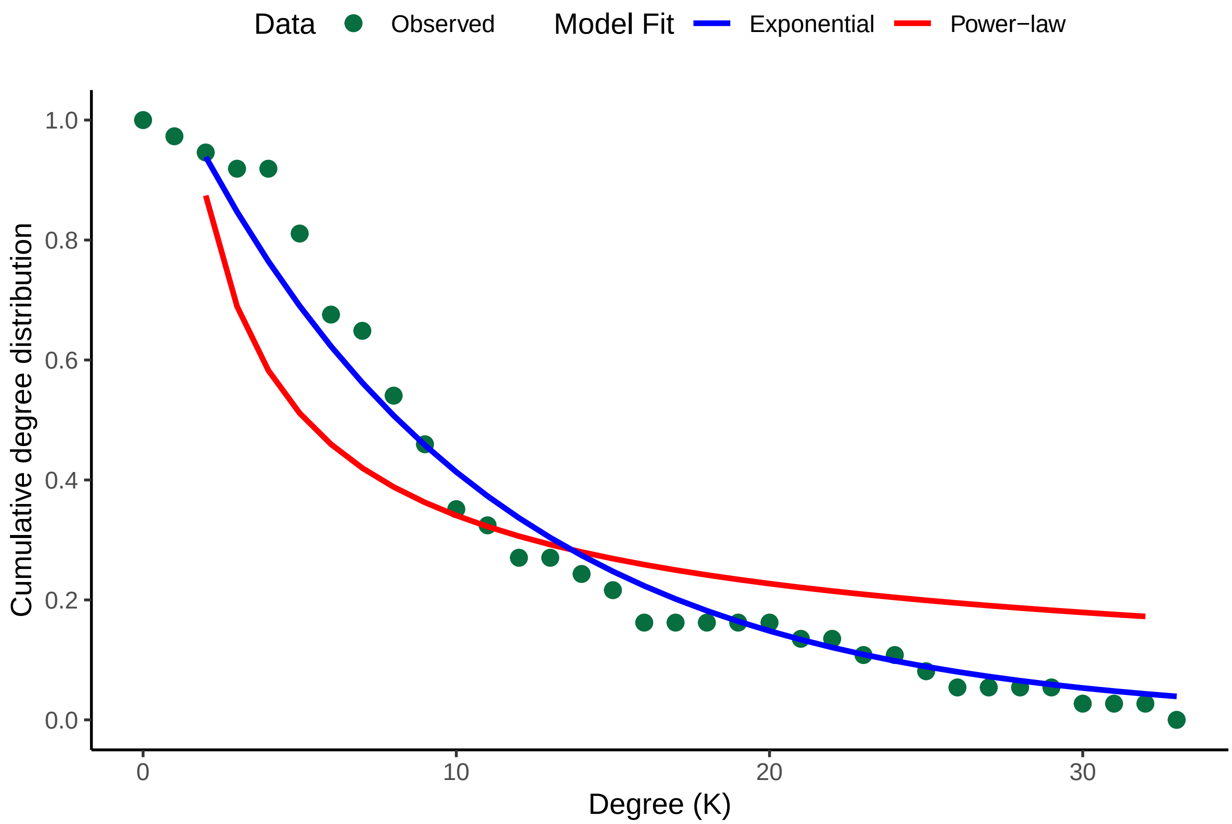

Figure 1: Cumulative degree distribution of the metaweb. The green points represent the cumulative degree of the species in Potter Cove, the blue line the exponential, and the red line the power-law model. The best fit was the exponential model.

References:

Corcoran, D., Ávila-Thieme, M.I., Valdovinos, F.S., Navarrete, S.A., Marquet, P.A., Kusch, E., 2023. NetworkExtinction: Extinction Simulation in Ecological Networks.

Supplementary 5: Module Roles

Modularity was calculated with an algorithm (Csardi and Nepusz, 2006) and then topological roles were assigned to the species (Figure 2). Topological roles were defined based on the standardized within-module degree and the among-module connectivity participation coefficient. The within-module degree is a relative measure of connectance of a species to other species within the same module, and the threshold for this measure is 2.5. The among-module connectivity is a relative measure of the amount of links from a species to other modules; here the threshold value is 0.625. The four topological roles are (i) Module Hub, with high within-module degree but low among-module connectivity, (ii) Network Connector, with high within-module degree and high among-module connectivity, (iii) Module Specialist, with low within-module degree and low among-module connectivit, and (iv) Moduel Connector, with low within-module degree but high among-module connectivity (Kortsch et al., 2015).

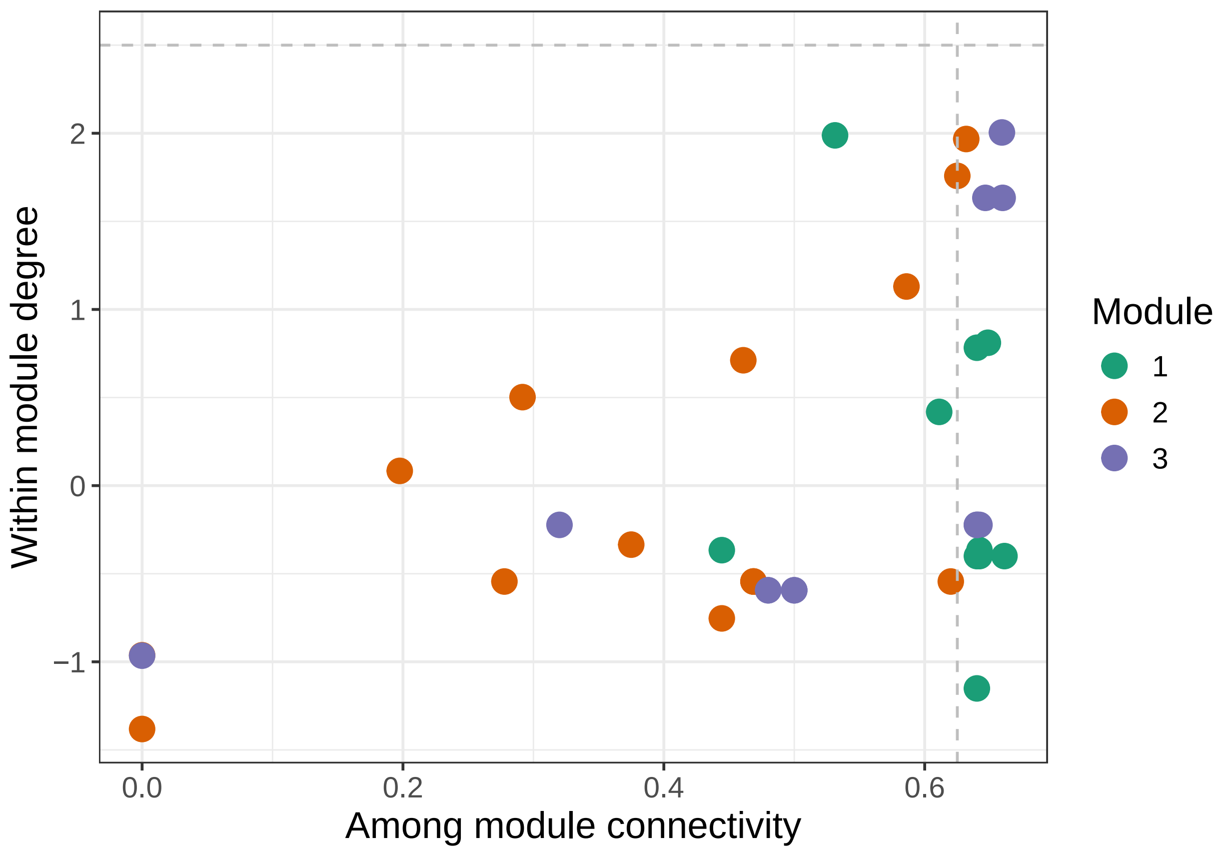

Figure 2: Topological roles of the species in the metaweb. The graph shows the within-module degree and among-module connectivity of the species, their module participation and which topological role they are assigned. The modules are indicated by the different colors and the species and exact values can be found in the table in Supplementary 3.

References:

Csardi, G., Nepusz, T., 2006. The igraph software package for complex network research.

Kortsch, S., Primicerio, R., Fossheim, M., Dolgov, A.V., Aschan, M., 2015. Climate change alters the structure of arctic marine food webs due to poleward shifts of boreal generalists. Proc. R. Soc. B Biol. Sci. 282, 20151546. https://doi.org/10.1098/rspb.2015.1546

Supplementary 6: Results of the Scenarios

Tables 5 and 6 consist of the results on the species level for the analyses of the scenarios and include the trophic level (TL), the total interaction strength (Tot_IS), the interaction strength for in- and out-interactions (In_IS, Out_IS respectively), and the total degree (Tot_Deg).

Table 5: Results of the bloom year ordered by trophic level

Species	TL	Tot_IS	In_IS	Out_IS	Tot_Deg
Clione antarctica	3.00	0e+00	0e+00	0e+00	1
Gyrodinium spp.	2.98	0e+00	0e+00	0e+00	14
Protoperidinium spp.	2.98	0e+00	0e+00	0e+00	10
Chaetognatha	2.84	0e+00	0e+00	0e+00	3
Nauplii	2.21	0e+00	0e+00	0e+00	10
Hydromedusae	2.16	2e-07	2e-07	0e+00	6
Ostracoda	2.16	1e-07	1e-07	0e+00	8
Calanoida	2.15	2e-07	2e-07	0e+00	20
Amphipoda	2.11	2e-07	2e-07	0e+00	9
Aloricate Ciliate	2.07	0e+00	0e+00	0e+00	8
Tintinnid	2.07	0e+00	0e+00	0e+00	9
Cyclopoida	2.00	1e-07	1e-07	0e+00	14
Limacina antarctica	2.00	0e+00	0e+00	0e+00	8
Meroplankton	2.00	1e-07	1e-07	0e+00	12
Harpacticoida	2.00	0e+00	0e+00	0e+00	1
Euphausida	1.82	4e-07	2e-07	2e-07	21
Benthic/epiphytic	1.00	0e+00	0e+00	0e+00	10
Cryptophytes	1.00	0e+00	0e+00	0e+00	11
Naviculoids	1.00	0e+00	0e+00	0e+00	9
Small naked dinoflagellates	1.00	0e+00	0e+00	0e+00	6
Detritus	1.00	9e-07	0e+00	9e-07	7
Large Thalassiosiroids	1.00	0e+00	0e+00	0e+00	7

Table 6: Results of the non-bloom year ordered by trophic level

Species	TL	Tot_IS	In_IS	Out_IS	Tot_Deg
Nauplii	2.59	0e+00	0e+00	0e+00	18
Protoperidinium spp.	2.27	0e+00	0e+00	0e+00	11
Gyrodinium spp.	2.16	0e+00	0e+00	0e+00	15
Aloricate Ciliate	2.04	0e+00	0e+00	0e+00	13
Tintinnid	2.04	0e+00	0e+00	0e+00	14
Amphipoda	2.00	2e-07	2e-07	0e+00	6

(continued)

Species	TL	Tot_IS	In_IS	Out_IS	Tot_Deg
Calanoida	2.00	1e-07	1e-07	0e+00	21
Cyclopoida	2.00	1e-07	1e-07	0e+00	17
Meroplankton	2.00	1e-07	1e-07	0e+00	20
Harpacticoida	2.00	0e+00	0e+00	0e+00	4
Ostracoda	2.00	1e-07	1e-07	0e+00	7
Salpida	2.00	2e-07	2e-07	0e+00	15
Benthic/epiphytic	1.00	0e+00	0e+00	0e+00	8
Chaetoceros spp.	1.00	0e+00	0e+00	0e+00	8
Cryptophytes	1.00	0e+00	0e+00	0e+00	9
Fragilariopsis spp.	1.00	0e+00	0e+00	0e+00	8
Naviculoids	1.00	0e+00	0e+00	0e+00	8
Other Phytoflagellates	1.00	0e+00	0e+00	0e+00	8
Pseudogomphoner	1.00	0e+00	0e+00	0e+00	4
Small Thalassiosiroids	1.00	0e+00	0e+00	0e+00	8
Small naked dinoflagellates	1.00	0e+00	0e+00	0e+00	6
Detritus	1.00	8e-07	0e+00	8e-07	6
Corethron pennatum	1.00	0e+00	0e+00	0e+00	6
Large Thalassiosiroids	1.00	0e+00	0e+00	0e+00	6
Pseudo-nitzschia spp.	1.00	0e+00	0e+00	0e+00	4
Other dinoflagellates	1.00	0e+00	0e+00	0e+00	2